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MILITARY STANDARDIZATION HANDBOOK

FUNCTIONALLY ORIENTED MAINTENANCE MANUAL (FOMM) WRITER'S GUIDE FOR MIL-M-24100B

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20301

MIL-HDBK-242
FUNCTIONALLY ORIENTED MAINTENANCE
MANUAL (FOMM) WRITER'S GUIDE FOR
MIL-M-24100B

1. This standardization handbook was developed by the Department of Defense in accordance with established procedure.
2. This publication was approved on 25 October 1974 for printing and inclusion in the military standardization handbook series.
3. This document provides guidance for the preparation of a Functionally Oriented Maintenance Manual. The handbook is not intended to be referenced in procurement documents except for guidance purposes, nor shall it supersede any specification requirements.
4. Every effort has been made to reflect the latest information on the preparation of a Functionally Oriented Maintenance Manual. It is the intent to review this handbook periodically to ensure its completeness and currency. Users of this document are encouraged to report any errors discovered and any recommendations for changes or inclusions to the Commander, Naval Ship Engineering Center, Department of the Navy, Center Building, Prince Georges Center, Hyattsville, Maryland, 20782.

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FOREWORD

The object of this handbook is to describe the Functionally Oriented Maintenance Manual (FOMM), specified in MIL-M-24100B, so that its concept and relationship to conventional manuals is fully understood in order that a technical writer experienced with writing conventional manuals can prepare the FOMM manual with the least difficulty and expense. To accomplish this, an in-depth study of the methods used for efficient and effective FOMM manual preparation is described herein.

Whenever any technical manual is prepared to support a weapon system or equipment, certain information is required to be included in the manual – usually, that information necessary to become familiar with the system, install it, operate it, maintain it, and repair it. Therefore, all technical manuals must contain the same type of information. FOMM is no different in this respect. The difference lies in the method of data presentation.

Traditionally, technical manual writers have divided information by chapters or sections for the purpose of separating information into definable categories (e.g., general information, installation, operation, theory of operation, maintenance, troubleshooting, repair, and parts data). The data is normally oriented toward the construction of the equipment. As the complexity of systems and equipment increases, it becomes increasingly difficult to divide information into discrete elements based on hardware packaging, and still properly explain the operation of the equipment and provide for effective troubleshooting. For this reason, different schemes for technical manuals, and modifications to requirements for conventional manuals, were developed to reorganize the information presented and to emphasize the "functional operation" of the equipment or systems. This reorganization was based on the premise that it is more important to know how the equipment functions than how it is physically packaged. Using the FOMM approach, it is easy to relate functionally presented circuits or components to the hardware items in which they are located.

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INTRODUCTION

FOMM is a technical manual concept designed to incorporate the best features of sectionalized types of manuals with those of entirely integrated manuals. In addition, it includes new features which overcome many of the inherent deficiencies associated with existing manual types, and makes the FOMM manual microform-compatible.

This is done by separating the information into two volumes. The information required to become familiar with an equipment or system, install it, operate it and maintain it, is best presented in the sectionalized format (Support Volume). The information required to troubleshoot and repair the equipment or system is best presented in an integrated format (troubleshooting/repair volume).

The Troubleshooting/Repair (T/R) Volume is the heart of FOMM. It contains all of the basic drawings, text, troubleshooting charts, parts-location diagrams, alignment, disassembly, reassembly and repair data necessary to effectively troubleshoot and repair the equipment. It is intended that the T/R volume stand alone so that a qualified technician can expeditiously isolate and correct a fault or casualty without reference to other documents. The support volume provides information for training, installation, operation, and maintenance.

Because of the importance of the T/R volume, and because it includes highly disciplined requirements and represents the highest percentage of production costs, this Writer's Guide slants toward providing the guidance necessary for its preparation.

A FOMM manual, by its nature, is a highly visible and disciplined technical document whose structure makes it easy to trace signals and/or information through its various levels. This ease of traceability makes verification and checking of the manual relatively simple. Accordingly, missing or inconsistent data is likely to be detected. If such detection takes place after the manual has been submitted for review, the result will be unnecessary expense and delay caused by the needed reworking. The writer's careful planning and attention to the detailed specification requirements during development of the manual will simplify its organization later. This care will also reduce the chances for improper presentations and/or inadvertent omissions.

FOMM concept. The FOMM concept requires that the manual include descriptions of how a system/equipment operates by use of functional diagrams. These are supported by text and troubleshooting charts (called Maintenance Dependency Charts or MDC's) organized to provide rapid fault isolation. The diagrams depict how the system operates and what portions of the system are operating during various modes.

A FOMM manual presents information in three hierarchical levels,as follows: (1) the overall information level that shows and identifies the "functional" makeup (major functions) of the equipment or system, and provides that information necessary to determine whether or not each of the major equipment functions is operating properly; (2) the major function level that provides information necessary to isolate an improperly operating major function to an item of hardware; and (3) the hardware level that provides for each unit or assembly of hardware that information necessary to isolate a fault to a defective component (or a group of components), and effect a repair.

The information at each of the hierarchical levels is arranged in data packages. Each data package contains all the information required to perform a given troubleshooting task without reference to any other data package.

The troubleshooting philosophy requires that the technician begin at the overall equipment information level to perform and complete a troubleshooting task. This task will then lead him to a lower-level data package, where another task must be performed and completed. This procedure continues until the lowest level data package has been entered and the final troubleshooting and repair task has been completed. Refer to figure 1.

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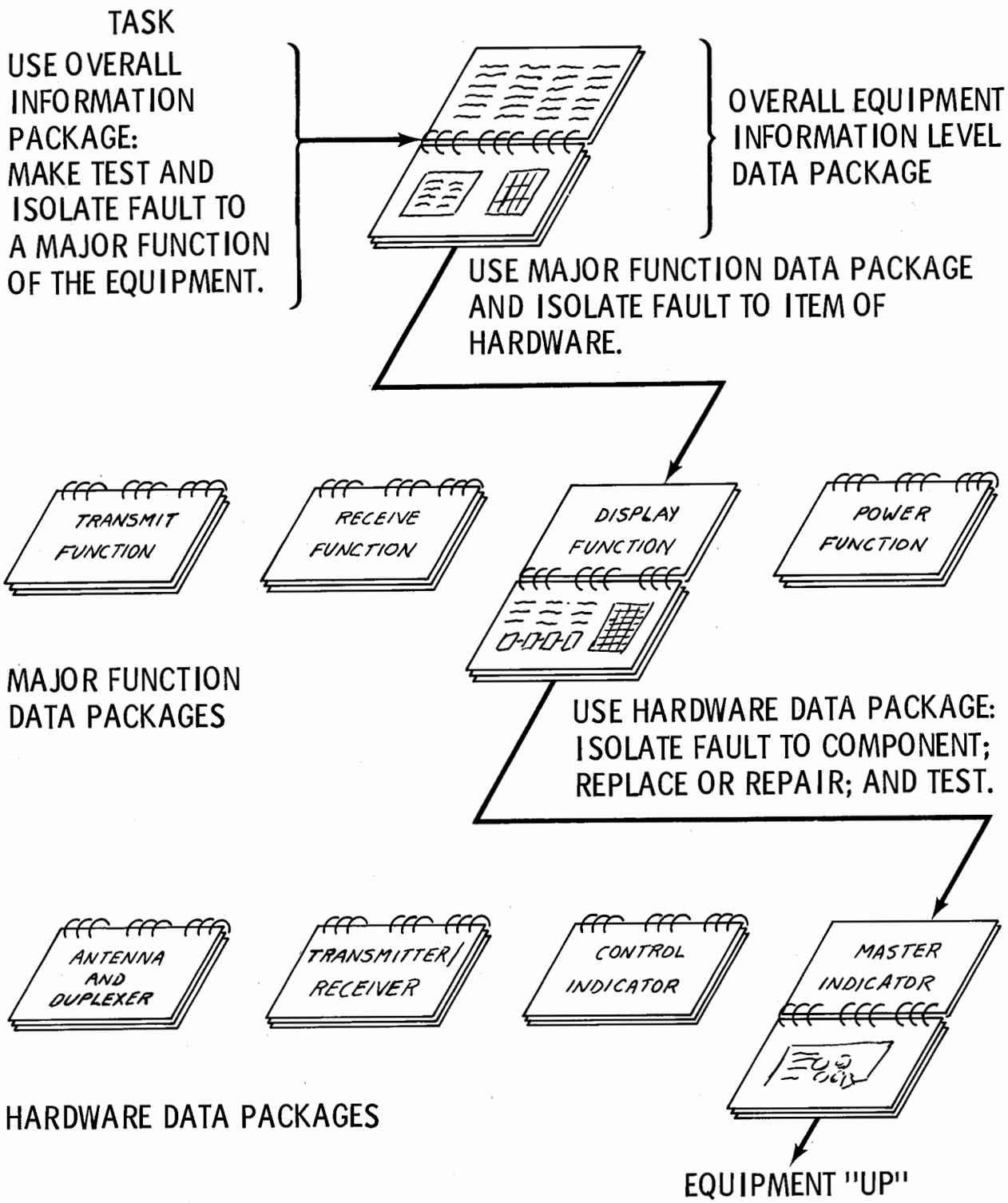


Figure 1. Troubleshooting Philosophy

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Data packages. Data packages comprise one or more illustration types and related charts and information. There are basically three general types of data packages, one for each of the three hierarchical levels previously described.

Overall function data package. There is one package of data for the overall equipment/system information. This data package identifies each of the major functions of the equipment or system and provides the information required to isolate any equipment malfunction to a given major function data package.

The overall function data package will consist of the following:

(a) One overall equipment/system functional block diagram identifying and showing the relationships of all major functions of the equipment.

(b) Keyed text explaining the operation of the equipment and, in general, how the major functions operate.

(c) An MDC which provides all the steps, tests, and signal specifications required to isolate a fault to a given major function.

Closely associated with the overall function data package, but not actually a part of it, are the following diagrams:

(a) A family tree that indicates the hardware items comprising the equipment or system.

(b) A cabling/piping diagram that shows the physical connections between major hardware units.

(c) A check-out chart that provides a quick method for testing the equipment and entering the overall or lower level MDC.

(d) Alignment information that is required for actions to be performed at the equipment level.

Overall equipment data packages may contain "intermediate level" data packages. These intermediate level data packages are expansions of the basic overall function data package, and are included only when required information cannot be properly arranged and included on a single sheet. All requirements that apply to the overall function data package apply to the intermediate level data packages. Typical use of an intermediate diagram is shown in figure 2.

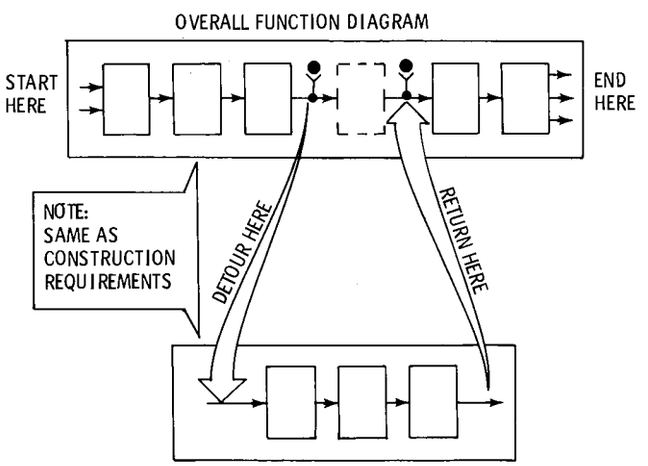


Figure 2. Intermediate Diagram

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Major function data package. There is a separate package of data for each major function of the equipment as identified in the overall data package. There are always two or more major function data packages, and they are normally arranged in the sequence shown on the overall function diagram. Each major function data package identifies the circuits and elements that are involved with the operation of the major function, and provides the information required to isolate any malfunction to a given hardware data package.

Major function level data packages consist of the following:

(a) One major function diagram.

(b) Keyed text explaining the details of the operation of the major function.

(c) An MDC providing all the steps, tests, and signal specifications required to isolate a fault to a particular item of hardware.

Hardware data package. There is a separate hardware data package for each unit, assembly, and subassembly. Hardware data packages are arranged in numerical or alpha-numerical sequence in accordance with designated nomenclature or identification numbers. Each hardware data package includes all the data necessary to locate and repair any malfunction that has been isolated to the hardware item.

Hardware level data packages will consist of the following for each unit, assembly, and subassembly, as applicable:

(a) A schematic diagram.

(b) Keyed text, explaining the function of the modules, circuits, and components.

(c) An MDC, providing all the steps, tests, and signal specifications required to isolate a fault to a particular circuit or component.

(d) A wiring diagram for front panels, chassis, and cabinets.

(e) A parts location illustration and data for locating and identifying all repairable and replaceable parts.

(f) Alignment and repair/overhaul procedures.

(g) Illustrated parts breakdown or exploded views.

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1. SCOPE

1.1 Writer's guide. The following information explains and amplifies the requirements specified in MIL-M-24100 and provides guidelines for planning and production of FOMM material. No deliberate attempt is made to state hard and fast rules. Rather, the discussion points out methods which have worked well in the past. Most writers who have delved into the FOMM concept have concluded that the production of a volume can be as much an art as it is a science and therefore some creativity is encouraged. This may be seen by checking the illustrations closely in this handbook. Practically every example shows some exercise of independent judgement of what constitutes the best and most effective method of presentation.

1.1.1 FOMM drawings, with the addition of zoned borders and title boxes, meet the requirements of MIL-D-1000, but it is a rare case when an engineering drawing prepared to the requirements of MIL-D-1000 meets FOMM requirements. This is because the requirements of MIL-D-1000 do not emphasize nor require circuits to be functionally arranged. The result is that the writer should spend much of his time in considering the production of schematics and block diagrams which he intends to include in his manual. As a consequence, the manual production hints provided here emphasize the development of schematics and block diagrams and how they are used as a source for other manual material. The first part of this handbook discusses the general philosophy of FOMM as it is implemented in the T/R volume. The philosophy is that the material is divided into data packages. Each data package centers about a schematic or block diagram and each package contains information identifying the next step the manual user should take. After this philosophy is explained, the methods of developing individual data items are discussed, and again it is seen that the schematics and block diagrams are used to guide the way.

1.1.2 After all portions of the troubleshooting/repair information are discussed, attention is turned to the support volume. Hints on preparing support volume material are limited to brief comments on required content, since the volume is a standard technical manual and production should not prove difficult to any writer skilled in his trade.

1.1.3 The handbook concludes with some points concerning estimating and bidding. This material is aimed at the supervisor who is approaching his first FOMM project, and whose options in developing the most efficient production methods are no doubt limited by company past practices and the present work loads. Unfortunately, these constraints may result in a marginal quality or an over-budget manual. Hopefully, the hints given will prevent this from being the case. After an organization has completed its first FOMM project, procedures for bidding and estimating should have been compiled, making reference to this section unnecessary.

1.2 FOMM manual. In the development of most technical manuals, all too often there is usually one concerned group that does not have a voice in manual content. This group consists of the ultimate users; the technicians who must use the manual as a source for preventive and corrective maintenance information. In some cases those who have had a voice in the contents have seemed to reorient the purpose of the manual. They have used the manual to point out the design concepts, provide a history of equipment development for the project office, include a checklist of contents for the reviewers, and serve as a convenient listing of federal supply numbers for a depot. Hopefully, somewhere in this mass of material, there will be information useful to the technician who must repair his equipment. The FOMM concept is one attempt to focus the writer's attention on the needs of the technician, and to ensure that all the information necessary for understanding operation and maintenance is included.

1.2.1 There is little, if anything, which is unique to the FOMM concept. Also, there is very little in the content of the FOMM which cannot, or should not, be included in a manual governed by a different specification. Perhaps an exception is use of color or shading, and even this factor is not universally prohibited by other specifications. The item which makes FOMM different from other manuals is that the specification is so precise in defining the rules of composition and the order and arrangement in which data is presented. Violation of the rules usually results in errors of omission or commission.

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1.2.2 Basic to the FOMM philosophy is that one picture is worth a thousand words. Therefore, the T/R volume has a maximum of graphs, charts, and diagrams with a minimum supporting text. This approach has one subtle advantage in ensuring that usable material appears in the manual. This involves the expected technical comprehension of the reader. Presently, the target is toward an individual with a ninth grade reading ability and limited technical training. With these guidelines, it is completely beyond the capabilities of most writers, editors, and reviewers to determine if the technical writing can be understood by the intended reader. However, there is some assurance of proper depth of coverage when the material is placed in pictorial form. This is based on the conclusion that if diagrammatic data is presented in the technical manual "as it appears in a textbook" then, by graduation from an appropriate course of instruction, the technician has demonstrated his ability to comprehend the pictorial material. Indeed, this concept has led some to conclude that the limited text in a FOMM is the key to comprehension.

1.2.3 A second factor in the FOMM concept is that the technician, while performing a troubleshooting problem, should work from the overall operational concept down to isolation of a failed component. The manual is thus arranged from the most general information (equipment/system level) in the front of the volume to the most detailed (hardware level) located in the rear of the volume. This troubleshooting philosophy is further promoted by uni-directional reference material, with sufficient information to follow a path from general information to more detailed data.

1.2.4 A third factor which has been closely identified with this type of maintenance information is the inclusion of troubleshooting data in the form of an MDC. MIL-M-24100 is not the only technical manual specification requiring MDC's, and there appears to be a trend for the requirement of this troubleshooting tool in more and more technical manual specifications. However, the FOMM concept requires that the schematics and block diagrams delineate functional divisions of the hardware and incorporate the functional division names into the heading of the MDC. At present, the Maintenance Dependency Chart is a somewhat maligned concept, probably because people expect maintenance personnel to use the chart when it is not necessary. Also, a good chart requires more effort to produce than troubleshooting information in a conventional format. An MDC is a useful tool when used for its intended purpose, which is to supply all the information needed by a technician when a fault is not easily isolated by a review of the block and schematic diagrams. It is expected that an experienced person will rely on the diagram, using the accompanying MDC as a guide to establish the direction in which troubleshooting will proceed. Further, it is expected that a qualified technician will use the chart only when he is dead-ended. The MDC also serves some peripheral purposes. Preparation of a Maintenance Dependency Chart, and checking it against operating equipment,should provide the writer with more than enough data for preparing text. Also, the compilation of one-sentence statements for each dependency line in the chart should result in an adequate outline for a topic in the instructor's guide for the equipment. If this procedure results in a poorly organized instructor's guide, then it is quite probable that the MDC is inadequate for its intended purpose.

1.2.5 For many who are associated with a FOMM project, the main question is: "What will it cost"? There is little doubt that, without competent personnel and proper organization, a FOMM will tend to cost more than a standard technical manual. However, since FOMM and conventional manuals should contain the same technical content presented differently, there should be little cost difference, even though MDC's are expensive items. It can be assumed, though, that alternate troubleshooting data with the same effectivity would cost as much or more. These life cycle costs are nebulous, but it can be pointed out that care in production of a FOMM should result in minimum change to a volume, and the visibility provided should aid engineering in designing equipment modifications. This is especially true when, over a period of time, the original designers are no longer available and the new responsible engineers must learn the present status of the design. Printing costs for a FOMM should be less than a conventional manual since normally there will be fewer pages involved. The use of color is an added expense since two press runs are required for some pages. By contrast, other approaches utilizing more than one color have required up to seven or more press runs.

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2. REFERENCED DOCUMENTS

2.1 The issues of the following documents, in effect on the date of invitation for bids, form a part of this handbook to the extent specified herein.

GOVERNMENTAL

SPECIFICATIONS

MIL-D-1000	-	Drawings, Engineering and Associated Lists
MIL-M-24100	-	Manuals, Technical: Functionally Oriented Maintenance Manuals (FOMM) for Equipment and Systems
MIL-M-38784	-	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	-	Printing Production of Technical Manuals, General Requirements for

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. TROUBLESHOOTING/REPAIR VOLUME

3.1 Scope and content. The troubleshooting/repair volume is described first because, in essence, it is the heart of FOMM. All information for a qualified technician familiar with the documented equipment is contained in this volume. The troubleshooting/repair volume provides information to fault isolate, repair, and align the equipment. Depth of coverage is determined by the echelon of repair designated, by the maintenance philosophy specified in the contract, and by training considerations. Most of the information in the troubleshooting/repair volume is an expansion of the data appearing in engineering documentation. It is augmented by the writer's knowledge of both the equipment and the technical concepts involved. A major portion of the volume content is the result of the analysis and interpretation of source data to provide troubleshooting and repair information.

3.1.1 Arrangement philosophy. The rules for material arrangement attempt to implement the adage "a place for everything and everything in its place". The philosophy of arrangement is to place information in the same order as it will be needed by one who uses the manual while troubleshooting his equipment. This implies a troubleshooting discipline, and the adopted philosophy that the troubleshooter should progressively eliminate portions of the equipment as not containing the possible failure. Thus, the material in the manual is arranged from the general to the particular. It is expected that when the using technician makes a troubleshooting decision, the next bit of information needed will be located on a succeeding page. Also, the arrangement plan attempts to aid those who habitually scan through a manual looking for information, rather than using the index.

3.1.1.1 Arrangement of data within the volume. The concept specified to accomplish the arrangement is called division by hierarchical level. Using this concept, all information concerning the overall equipment is grouped together near the front of the volume. This is called the overall function information (hierarchical) level of data. Following the overall information is all data concerning functionalization of the equipment (or major function level). Finally, the last pages in the volume contain the most detailed or hardware level information (or hardware packages). The use of a particular portion of the volume for a given type of data is carried over to the introductory and general information. This introductory and general expository data precedes the overall equipment information.

3.1.1.2 Arrangement of introductory material. There are three types of introductory material, arranged as follows:

(a) Introduction to the manual
(b) Introduction to content use
(c) Introduction to the equipment

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The introduction to the manual covers such items as functional index, description of contents, adopted symbology, and change record. The content-use introduction includes an explanation of how the manual is organized and the relationship of diagrams to each other, instructions for interpreting the MDC's, a compilation of truth tables and logic diagrams for two-state devices, a glossary of terms, etc. The equipment introduction consists of interconnection with other equipments, a description of the hardware, the division of hardware into assemblies, and the necessary check and calibration procedures required to determine if the equipment is operating properly. The equipment introduction is included in the overall information data package as shown in chart 1 of MIL-M-24100B.

3.1.1.2.1 The specification allows some latitude in the arrangement of introduction material in order to reduce the possibility of blank pages. However, in this section, as in the manual as a whole, related information which can be placed on facing pages should not be separated merely to reduce the number of blank pages.

3.1.1.3 Arrangement within a hierarchical level. The rationale for arranging material by hierarchical level is to aid maintenance personnel in understanding the equipment and in accomplishing their tasks. The same reasoning is used in the arrangement of material within a hierarchical level, but the method of accomplishment is different. The concept applied is called arrangement by data packages, and the objective is to present all the necessary information for a particular troubleshooting/repair operation at one easily located spot in the manual. Thus, at the "hardware" hierarchical level, a data package contains all the detailed information for one unique hardware assembly. The second hierarchical level of data packaging contains all the fault isolation information for one major functional division. At the overall hierarchical level there is only one data package. However, page size limitations may necessitate several pages in the packages.

3.1.1.3.1 Sequencing. The rules for the sequence in which data packages are arranged is a function of the hierarchical level. For major function information, the major function data packages, each consisting of a major function diagram and related subfunction diagram and MDC's, are arranged according to the keyed text numbers as shown on the overall function diagram. Hardware data packages are arranged in sequence, using the reference designation or manufacturer's part number as a guide. Intermediate and subfunction information is treated as part of a larger package, and placed immediately after the applicable overall or major function package.

3.1.1.3.2 Arrangement within data packages. Rules for arrangement of material within a data package must be flexible in order to allow for variations in page size, method of binding, and the amount of required information. Basically, it is desirable to have all related information on one page, or two facing pages. These rules are discussed later.

3.1.2 Content of a data package. In general, the format of data packages remains the same from one hierarchical level to another. In particular, it is to be expected that all packages, as a minimum, should contain a diagram, text, and MDC with signal specifications. In the FOMM concept, the diagram is meant to be the center of attention. All other information is presented as an extention in support of the diagram. As an example, the text is keyed by number call-out (key number) to each functional grouping on the diagram, and is written to the level of detail depicted on the diagram. Likewise, the MDC identifies each functional grouping on the diagram. The signal specifications are a record of voltage levels or waveshapes present at the interfaces between functional divisions. Other than the detail of coverage, the difference between the contents of a data package at each hierarchical level is in the type of additional material provided. At the hardware level, there is repair/alignment information. At the major function level there may be a discussion of how this portion of the equipment contributes to the overall mission. At the overall level, system alignment and any additional overall information required for corrective maintenance is included.

3.1.2.1 Overall function data package. This data package provides a functional overview of the entire equipment. It contains the overall diagram, the overall text, and the overall MDC. Figure 3 shows the arrangement of the data in an 11 x 27 inch format. This data package is augmented by the family tree, the cabling diagram, and any other information, such as equipment alignment data, that is required to understand the overall equipment operation and to isolate a fault to a major function.

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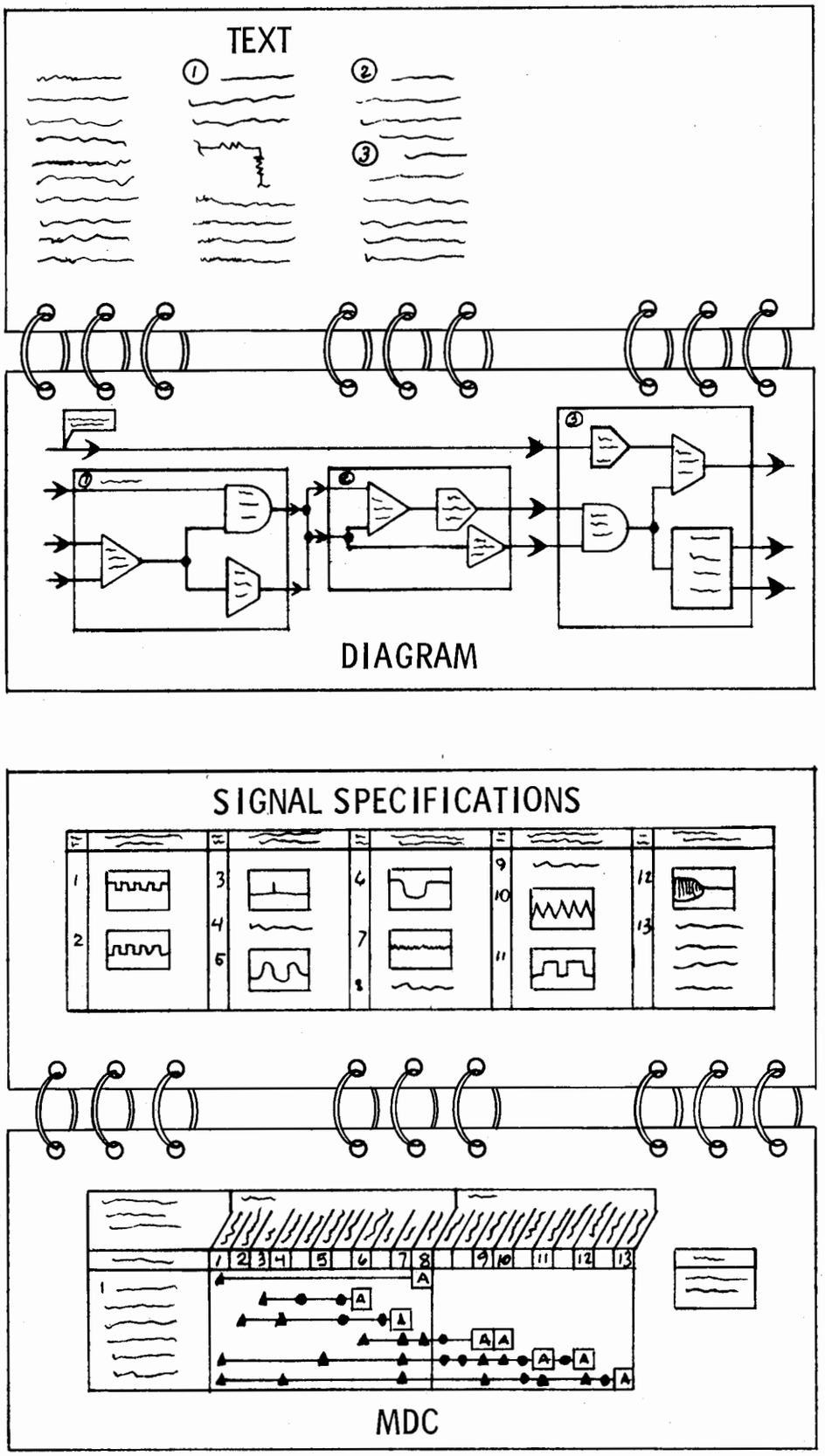


Figure 3. Data Package Arrangement-Overall and Major Function

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3.1.2.1.1 Overall diagram. This diagram identifies each of the major functions of the equipment and depicts the flow of information between major functions. It also shows data and power connections to and from external equipment. Each signal line is identified by the same name and usually by the same type of arrowhead as that shown on the major function diagrams. All front panel nomenclature and controls shown on major function diagrams are repeated on the overall diagram, within the block representing that major function. When an overall diagram becomes too large to fit on one page, an intermediate diagram(s) must be generated. Additionally, the overall diagram indicates the "major function" data packages that are included in the T/R Volume; that is, there will be one major function data package for each major function identified on the overall diagram. Note: When preparing the overall diagram, the objective should be to minimize the number of functions consistent with the complexity of the equipment or system.

3.1.2.1.2 Overall text. The text for the overall diagram is located on the same or facing page as the diagram. It contains two parts: a description of the equipment operation which pertains to the entire diagram, and keyed paragraphs explaining the purpose and operation of each major function. The purpose of the text is threefold: to explain what the equipment does, how it functions, and how each of the major functions contribute to the equipment mission.

3.1.2.1.3 Overall MDC. The MDC for the overall diagram must provide that information required for fault isolation to a major function. The preparation of the MDC is in accordance with paragraph 3.7 of MIL-M-24100.

3.1.2.1.4 Intermediate diagram. An intermediate diagram is an expansion of the overall diagram (because there is insufficient space to show all of the major functions) and illustrates several interrelated major functions. Intermediate diagrams are similar in presentation to an overall diagram. One or more intermediate diagrams may be generated if page size limitations prevent showing all individual major functions on a single overall diagram. Like the overall diagram, the intermediate diagram should contain all applicable front panel nomenclature, and interfaces to other major (or intermediate) functions. It must be remembered that all of the input and output interfaces must be identically identified and traceable from the overall diagram from which they were expended. Text supports the diagram in a manner similar to the overall text. In some cases it may be advantageous to produce an MDC to accompany the diagram. Factors influencing that decision would be the size and complexity of the overall MDC, and the equipment maintenance philosophy.

3.1.2.1.5 Additional data items. Four items related to, but not necessarily included in, the overall data package are:

(a) The family tree diagram
(b) The interconnecting cabling/piping diagram
(c) The equipment alignment
(d) The turn-on/check-out chart

For equipment requiring large troubleshooting/repair volumes these items precede the overall data package, and are arranged in the order shown. For less complicated equipment these items may be integrated with the overall data package.

3.1.2.2 Major function data package. A major function data package is to be developed for each major function identified on the overall function diagram. Each major function data package contains three related items that explain and provide troubleshooting information for a major function of the equipment. These items are the major function diagram, keyed text, and MDC. A major function is a practical grouping of related subfunctions performing a separable and definable task. Decisions as to what actually constitutes a separable task should be made with regard to the equipment being documented. The decisions involved do not lend themselves to hard and fast rules because of the variations in the type of information to be considered. The functionalization adopted by design engineering can be a good starting point. One indication of effective functionalization is the interfacing between major functions. The best arrangement is usually the one that results in the fewest interfaces. Irrespective of this, the selection must ultimately agree with the major function identified on the overall diagram, and of course it is subject to trade-off consideration.

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3.1.2.2.1 Major function diagram. The major function diagram is a symbolic representation of a major function. The diagram should show hardware identification, interconnections, and electronic/mechanical interaction between subfunctional units. If a major function exceeds page size limitations, subfunction diagrams must be generated. Subfunction diagrams are to have inputs and outputs that are identically identified and directly traceable to the major function diagram from which they were derived.

3.1.2.2.2 Major function text. The major function text presents information in two parts. The first part consists of the opening paragraphs, which explain what the function is and what it does relative to equipment operation. The second part is presented in paragraphs keyed to functional groupings on the diagram, explaining their contribution to the major function. When it becomes necessary to generate a subfunction diagram, the major function text must discuss the operation of subfunction (in general terms) and how the subfunction relates to the major function.

3.1.2.2.3 Major function MDC. The major function MDC is directly correlated to the overall MDC. The purpose of the major function MDC is to provide information for isolating failures that have been determined to be within a major function to a hardware division or chassis mounted part. The standard MDC format is used. Procedure column instructions should track those of the overall MDC, with additional check-out steps added as required.

3.1.2.2.4 Subfunction data package. A subfunction consists of the functional elements making up the major function. If the number of major functional elements exceeds the number that can be arranged on one diagram, one or more subfunctions are removed and shown on separate diagrams. Diagrams are supported by keyed text and an MDC.

3.1.2.3 Hardware data package. A hardware data package documents one unique physical division of the equipment, and contains the following items as applicable:

(a) Schematic diagram
(b) Text
(c) MDC
(d) Repair/alignment procedures
(e) Wiring diagram
(f) Parts data

Figure 4 shows a sample arrangement of the hardware data package, but there is considerable flexibility in this area.

3.1.2.3.1 Hardware schematic. One hardware schematic diagram is prepared for each uniquely definable assembly and subassembly of the equipment. It illustrates the relationships between all of the components contained on the hardware. A subassembly can be shown schematically on an assembly schematic when this aids in understanding circuit operation. For example, power supplies frequently have power transistors and large components mounted on a heat sink assembly, with the controlling circuitry located on a printed circuit board subassembly. One schematic of the entire power supply provides a more usable diagram than two separate ones. Like all other diagrams, the number of assembly and subassembly diagrams that should be contained on one page is limited to the page size and arrangement of facing text.

3.1.2.3.2 Schematic text. The purpose of the schematic text is to explain the operation of the circuitry depicted by the schematic diagram. The information is presented in two parts: the opening paragraph(s) explain overall operation, and the numbered (keyed) paragraphs explain the operation of the functional entities of the diagram. In cases where there are no functional entities (mother board, backplane, etc.), there will be no text.

3.1.2.3.3 Hardware MDC. The hardware MDC is used for fault isolation down to a function entity (one or more parts), on one uniquely identifiable hardware. The hardware MDC is in standard MDC format. MDC's are not required for non-repairable modules.

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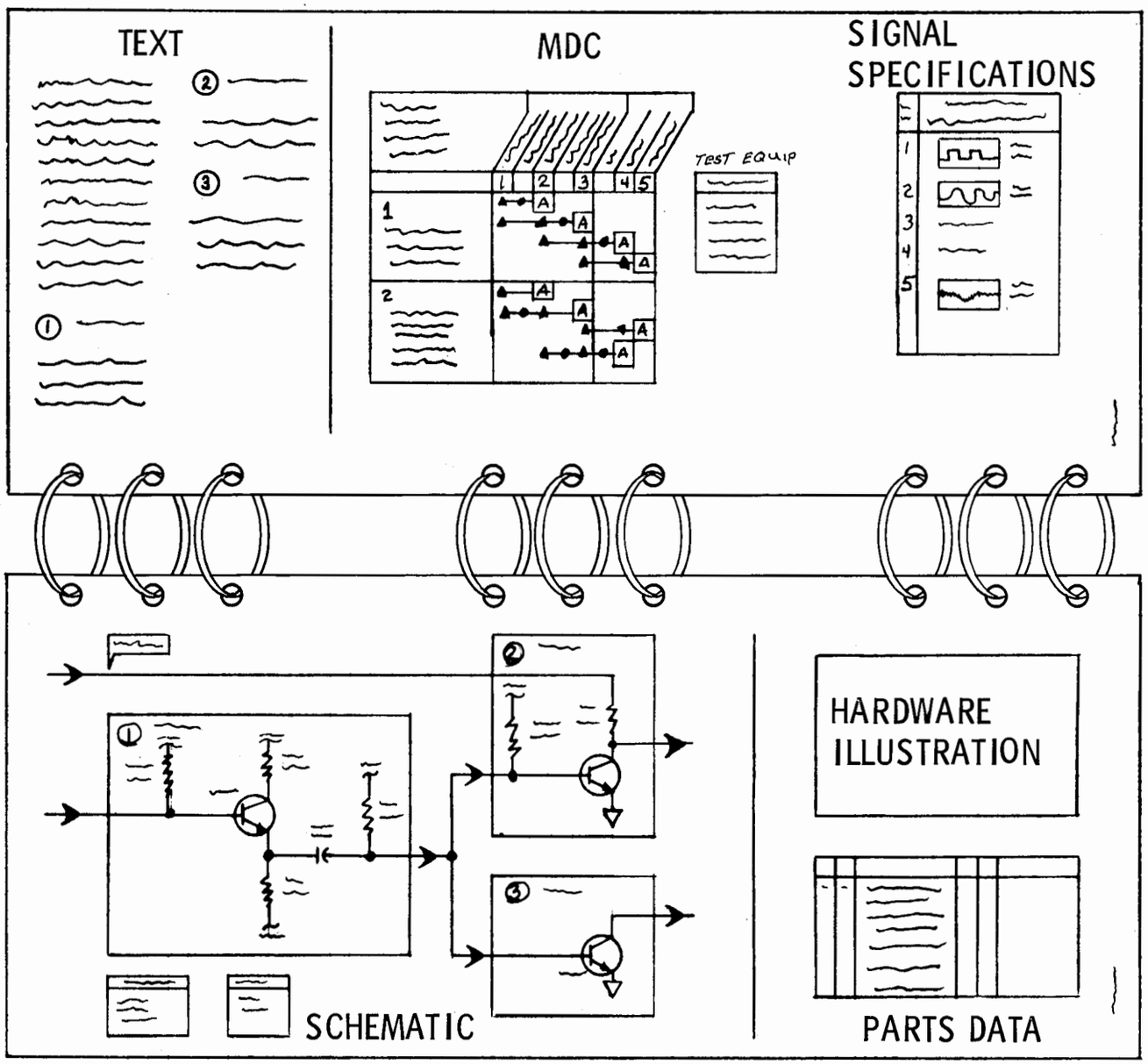


Figure 4. Hardware Data Package Arrangement

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3.1.2.3.4 Hardware repair/alignment instructions. The hardware alignment instructions describe the adjustments required to bring hardware outputs into tolerance. The alignment procedures shall include a list of required test equipment, and illustrations of any special test setups. The purpose of hardware repair instructions is to explain in detail the procedures required to access the hardware, to remove and replace it, and/or to repair it. These instructions are usually associated with mechanical assemblies, but should be included when access to any assembly is not obvious. Exploded views and other assembly drawings should be used to support these procedures. A list of special tools or equipment required and an exploded view or other assembly diagram shall be included. The exploded view can be used as the parts data illustration, providing it satisfies the requirement for parts location of all parts and components on the assembly.

3.1.2.3.5 Parts data. The hardware parts data provides a means of locating and identifying piece parts and special hardware items. The parts data consists of the parts location illustration and the parts data table. If a separate parts list is required by contract, the hardware package parts data table need not include ordering information (part number, manufacturer). Only the reference designators, generic names, and locating coordinates of the parts need be listed.

3.1.2.3.6 Wiring diagrams. Wiring diagrams illustrate point-to-point wiring on a chassis, backplane, or front panel. In the case of a backplane (or any other very densely wired panel), a double-entered wire run list may be substituted for the wiring diagram. Wiring diagrams should identify components, terminals, wire numbers, and color code.

3.1.2.4 Arrangement within a data package. Since a data package centers about the diagram, the remainder of the material (text, MDC, etc.) is located with respect to the diagram. The ideal condition for top-bound (large page size) volumes is to place all of the data for one package on one page, or on facing pages. For left-hand bound volumes the ideal condition is compromised in order to produce a more convenient size of volume. Consequently, the rules for data arrangement become a function of manual binding. Note that for any FOMM volume the writer must pay particular attention to arrangement, and assume some of the responsibilities which might otherwise be delegated to an editor.

3.1.2.4.1 Arrangement in top-bound volumes. A single sheet diagram is always placed on a bottom page so that this information is nearest the reader when the volume is open. Text describing diagram content is located on the same page as the diagram, or on the facing page. Within certain guidelines the remaining data may be placed where space permits. The guidelines are that exploded views remain with repair/alignment instructions, and signal specifications remain with the MDC. If the option exists, place the MDC on the same page as the diagram. This will help reduce the number of two-color pages and thus minimize production costs. When the MDC requires a full sheet, the preferred location is the next bottom page following the diagram, with signal specifications on the facing page. If signal specifications require more than one sheet, those which cannot be shown facing the MDC should be placed on the page(s) following the MDC. If it is desired to reduce blank pages, the MDC and signal specifications may be located prior to the diagram rather than following, but it is suggested that this arrangement be discussed with the customer prior to adoption. Multi-sheet diagrams should occur only with hardware data packages. If the diagram is contained on two sheets, the diagrams should face each other, providing that all text is also on the two sheets. For all other cases, the diagram sheets are located on sequential bottom pages.

3.1.2.4.2 Alternative arrangement for left hand bound 8½ x 11 inch volumes. (Note: This alternative must be specified in the contract or order.) The objective for this type binding is the adoption of a standard page size. This concept is a combination of standard size pages and variable size foldouts. Only diagrams, and possibly MDC's or exploded views, should be foldouts. The text should be the first item in the data package, followed by the diagram. The text should end on the page facing the diagram with the remainder of the page blank or used for material associated with the text, such as truth tables. If the text requires more than one but less than two page units, the diagram apron may be used. The apron of foldouts may also be used for such items as explanation of special test setups. The apron should not be used for information considered to be part of the diagram. For example, diagram notes are part of the diagram and should not appear on the apron. As is true for most technical publications, the back of a foldout should not normally be used. Signal specifications follow the diagram and precede the MDC, with any remaining data package items following in some logical sequence. This type of arrangement is the least disciplined method of organization and requires careful judgement in order to preserve the integrity of the data packages.

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3.2 Trackability. One of the major tasks for a troubleshooter is the tracking of a signal(s) through the divisions of an equipment until he locates the cause of a failure. In the accomplishment of this task, the maintenance man uses his technical manual to follow the signal path from diagram to diagram. In the FOMM concept, trackability is the adequacy of the information to permit tracking between different diagrams. In some cases, the path followed is from one diagram to another at the same hierarchical level. The adequacy of this transition data is called the horizontal trackability. If the path is from one hierarchical level to another, the adequacy of the transition data is called the vertical trackability. A diagram is considered completely trackable when it contains all the required information to support the optimum fault localization scheme. The inclusion of additional information which might aid a different troubleshooting concept is discouraged. As an example, it may be noted that on schematic diagrams the source and destination of signals are not shown, and the block diagram related to the circuit function is not identified. This is done to promote the most effective method of troubleshooting, i.e., the use of the upper hierarchical information for isolating faults to a hardware assembly. This troubleshooting concept of isolating from functional level to hardware level information also aids the manual writer in reducing costs and minimizing errors, since a diagram at one hierarchical level has no cross reference information to a higher level.

3.2.1 Rules of trackability. Trackability is obtained by the adoption of basic rules for consistency of presentation and special rules for interfacing between diagrams. The consistency rules assist in the recognition of identical information when the user moves from diagram to diagram. The interface rules control the tracking path. The applicability of the rules differs between vertical or horizontal trackability, and diagram types. That is, whether the track is from MDC to MDC, block diagram to schematic, or MDC to block diagram, etc. These rules are discussed in 3.2.2 and 3.2.3.

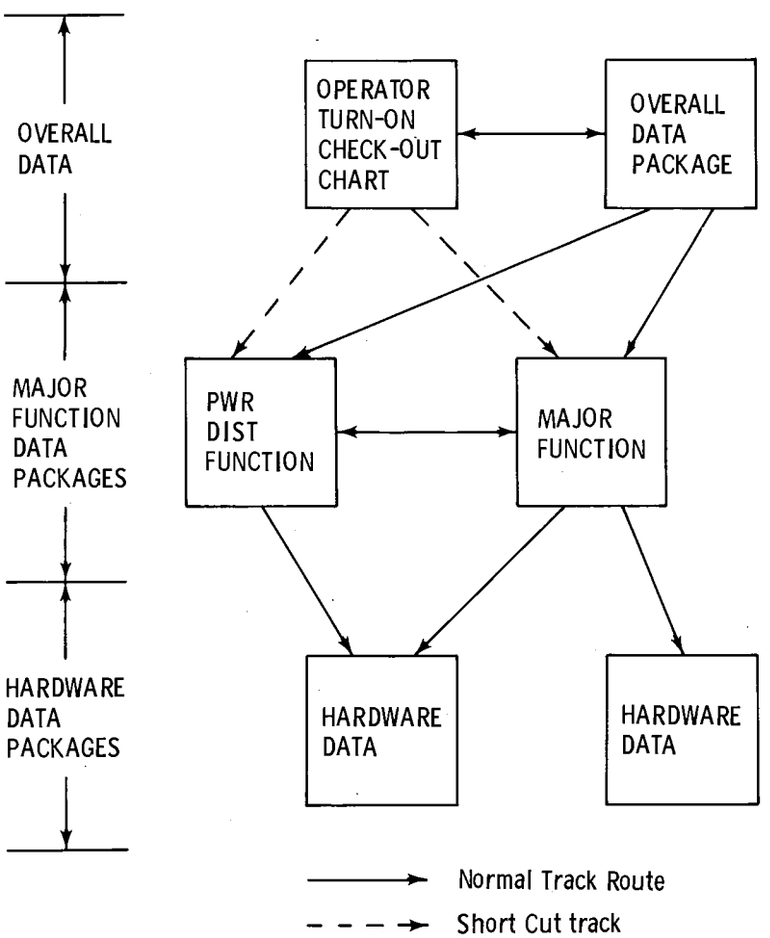


Figure 5. Tracking Route

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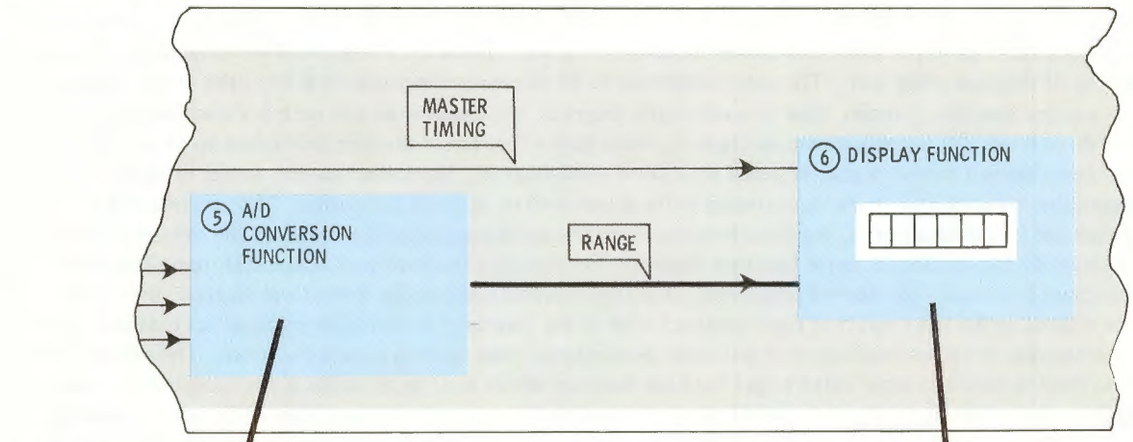
3.2.2 Vertical trackability. The primary concern for vertical trackability is the inclusion of sufficient information to allow tracking from an upper hierarchical level diagram to a lower. There are two general tracking paths, depending upon the type of diagram being used. The path considered to be of greatest importance is from the overall block diagram to a major function diagram, then to a schematic diagram. An example of this path is shown using the portions of three levels of diagrams shown on figure 6. Note that a functional division delineated on an upper level diagram is identified and shown in greater detail on a lower level diagram. Note also that the names of signals and, in most cases, line weights, and arrows are consistent from one level of diagram to another. Thus, in tracking a signal through the hierarchical levels, the troubleshooter uses the name appearing in a block on the overall function diagram to identify the applicable major function diagram. He then uses the hardware nomenclature on the major function diagram to identify the desired schematic. It is also expected that on the lower level diagram, the signal line will be located at the left (input) or right (output) edge of the diagram. It should be pointed out that reference is made to a diagram by its assigned name or reference designation; page number data is not given. The manual user may have to turn to the functional index to get the page number, which may be desirable if the equipment requires many pages for documentation. For less complicated equipments, the user should have little difficulty in locating the desired diagram by scanning through the volume. He should already know that the major function information follows the overall function diagram, and appears in the same sequence as the text numbers for the overall description. He should also be aware that the hardware information is last in the volume, and is arranged in reference designation or manufacturer's part number sequence. The second vertical tracking path is from the turn-on/check-out chart and the overall MDC, to a major function MDC, and finally to a hardware MDC. As is true with the block diagram to schematic path, the MDC path may detour to include intermediate or subfunction MDC's. Additionally, in the case of MDC data, it is possible to branch to alignment and other maintenance or troubleshooting instructions by placing references in the MDC to the particular instruction. Tracking from the turn-on/check-out chart to a lower level MDC is provided by including function names in the chart heading. The function names identify the MDC, showing detailed signal development data for the front panel indications included in the chart. The same technique is applied to the overall MDC, but in addition to front panel indications, the signals at functional interfaces are identified. Tracking from the major function to the hardware level is by inclusion of hardware identification and measurement point in the MDC heading. It is expected that the troubleshooter will use an MDC in conjunction with a block diagram, therefore the block diagram will be the primary source of vertical tracking data. For horizontal tracking, perhaps the only MDC's will be used.

3.2.3 Horizontal trackability. This type of trackability is only applicable to overall function and intermediate function diagrams and major and subfunction information. At the hardware level, the FOMM troubleshooting philosophy precludes the need for horizontal trackability. Some tracking information does occur, however, due to the consistency in the assignment of signal nomenclature. At the major and subfunction level of block diagrams, interfacing information is provided by starting (terminating) a signal line in a blue interface box containing the name of the major function which is the source (destination) of the signal. This is demonstrated in figure 7, along with additional means used for positive identification as follows:

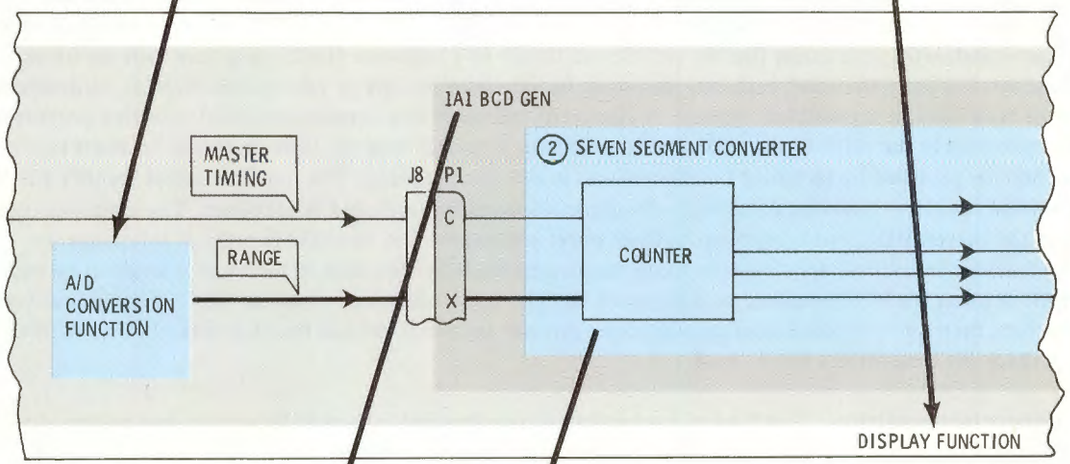
(a) Identical signal name on both diagrams.
(b) Compatibility of line weight and arrowhead.
(c) Both diagrams showing a common connection point identified with hardware reference designation. Examples of hardware shown for overlap are: plugs with pin identification, test points, and terminal boards with terminal identification.

The above method of providing horizontal trackability applies for functional signal lines. The interfacing of power is also important but, in order to prevent clutter, the interface is often implied rather than shown diagrammatically. A suggested method of showing implied interfaces is to use breaklines and voltage values where the voltage name corresponds to the name of the power distribution drawing showing the origin of the voltage. Horizontal trackability is provided in the heading of a major function MDC by identifying the source MDC for all inputs and the using MDC for all outputs. An exception is that the outputs shown on a power distribution MDC may be identified by distribution (measurement) point only, rather than the identification of all using functions. For both MDC's and block diagrams, locating the interfacing signal is aided by showing origination at the left hand side and destination at the right hand side.

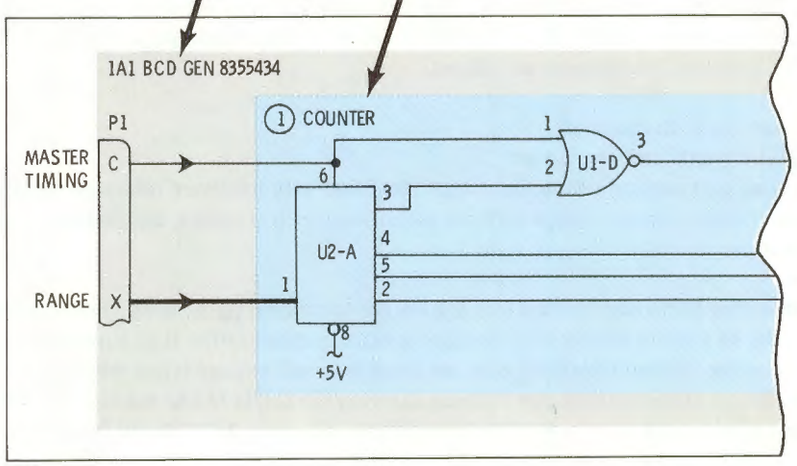
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a. OVERALL FUNCTION DIAGRAM



b. MAJOR FUNCTION DIAGRAM



c. SCHEMATIC DIAGRAM

Figure 6. Vertical Trackability Example

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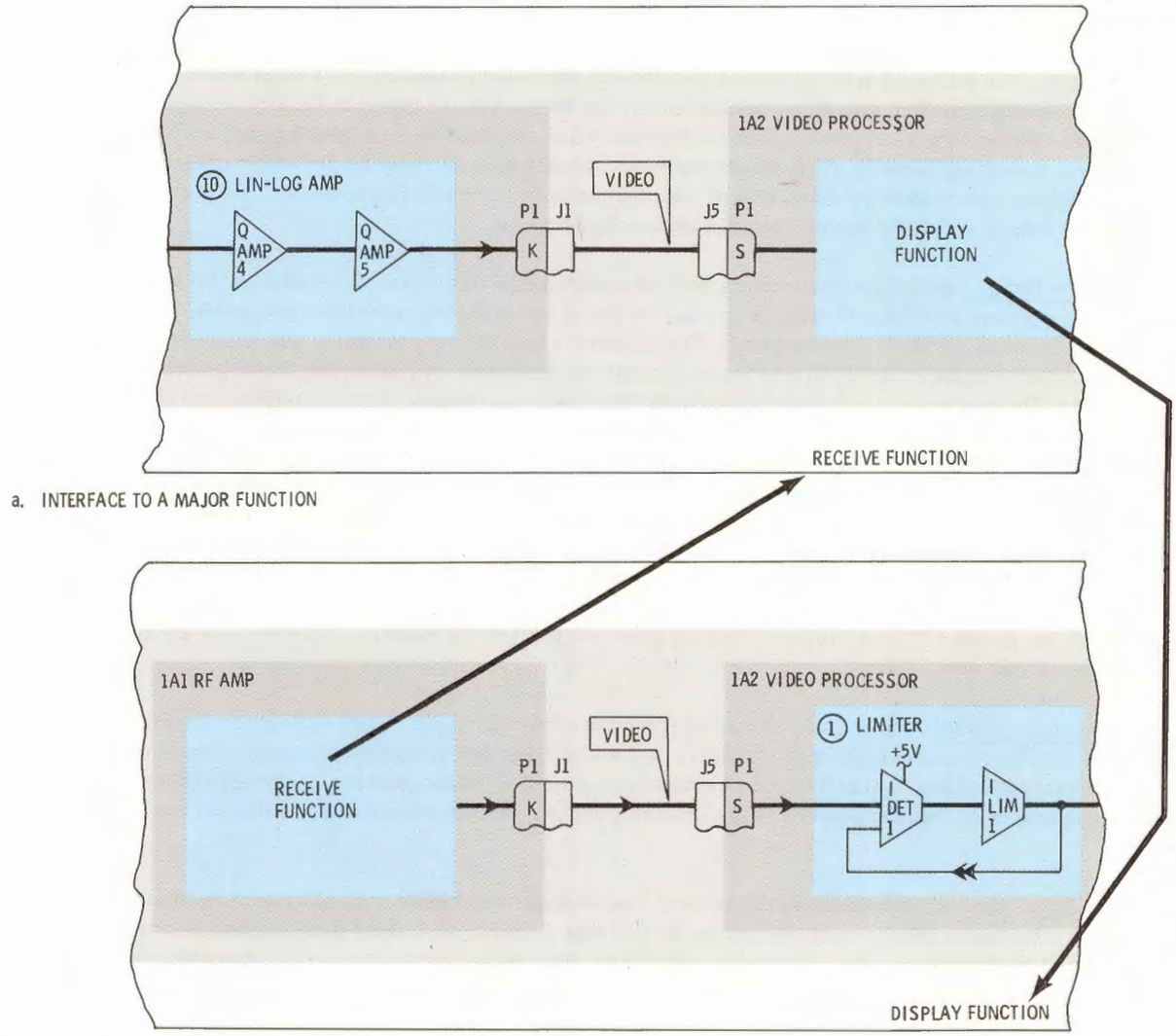


Figure 7. Horizontal Trackability Example

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3.3 FOMM techniques.

3.3.1 Introduction. The information in this section provides the writer with guidelines for FOMM manual preparation. Some of the concepts involved may be completely new to the writer, and the means to the required ends might not always be obvious. The techniques included in this handbook are meant to be suggestions and are in no way intended to, nor could they possibly, cover all contingencies. Their purpose is to enable the writer to make knowledgeable decisions concerning the presentation of the information for which he is responsible. The techniques are presented in the order in which the manual elements are usually prepared.

3.3.1.1 Microfilm format. Microfilm requirements must be considered in the preparation of all data (see figure 8). The information on any page over 8½ x 11 must be arranged so that it breaks cleanly into microfilm panels. Only horizontal lines can cross the divisions between panels. This is required because the information contained on the microfilm is read in frames equal to about 8½ x 11 inches and will be reproduced by a microfilm reader-printer on 8½ x 11 inch sheets. The user must be able to join the sheets cleanly with no overlap. Each microfilm panel should be completely identified with the manual publication number, page number, and division of page (e.g., 2-102a, 2-102b, 2-102c). Lower case letters should be used for division of page identification to avoid confusion if at a later date an "A" revision page must be inserted. The security classification must be indicated on every panel of a classified page. Also, the index for Volume 1 and the functional index of Volume 2 must indicate the microfilm/microfiche number and the frame number so that the technician can get the correct film and access the frame. It must be kept in mind that limitations of microfilm readers are such that the technician must scan back and forth when reading drawings over one frame and does not see the entire diagram during any given time. Even though microfilm reader-printers that can handle a 27 inch horizontal size are available, it cannot be assumed that every user will have access to a machine of that type.

3.3.2 Hardware data package. The techniques used in preparing a hardware data package will vary according to the type of equipment being documented. For example, a digital schematic and an electromechanical schematic require quite different techniques. A schematic of a mother board may have no components, and require no text or MDC. The maintenance philosophy adopted by the procuring activity will also influence the content of the data package.

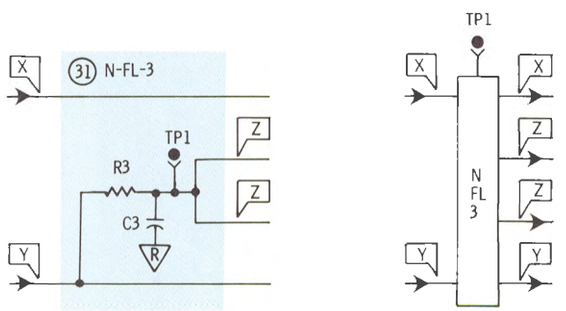
3.3.2.1 Schematics. Blocked schematics are the primary data elements of a FOMM manual. Their importance lies not only in the information they present, but also in the fact that decisions concerning their functional presentation directly affect other data items. The following is a list of rules that can be applied to blocked schematic diagrams:

(a) All circuit elements must be contained in functional entity boxes and all circuits presented in textbook form.
(b) Similar circuits must be laid out in the same form.
(c) Every functional entity must have a mnemonic identifier code.
(d) All inputs and outputs must have meaningful names.
(e) Signal, voltage, and mechanical breaklines must be identified.
(f) Signal lines must be properly identified (line weight, arrowheads, and signal names).
(g) All power must enter through C-PWR-1/C-PWR-2, etc.
(h) Adjustments, controls, and test points must be cut out of the blue shading and be shown on the grey level that corresponds to their proper hardware level.
(i) All components must be identified by their reference designation and type (e.g., 2N1325) or value (e.g., 100K) (see MIL-M-24100B, 3.3.2.6.1.7).
(j) Hardware reference designation and full name must appear in the upper left corner of hardware boundaries.
(k) The reference designator prefix, which should be placed in front of the individual circuit element reference designation to give its full reference designation, must appear on the lower right corner of the hardware boundaries.
(l) The source data block must contain the number(s) and revision letter(s) of all source engineering drawings which were used in the diagram preparation.

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(m) The general notes block must be complete.
(n) The flag note block must contain complete information concerning each flag note used on the drawing.
(o) Signal flow shall be left-to-right and top-to-bottom.

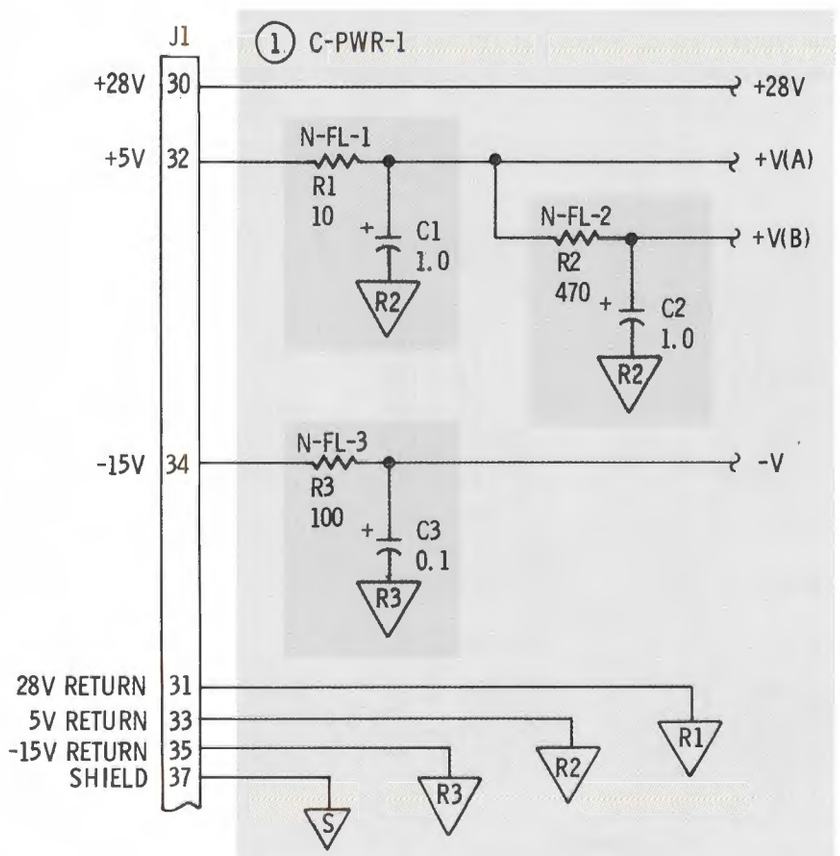
3.3.2.1.1 Functional entities, as defined in MIL-M-24100B, are groupings of parts connected together to perform a basic function (for example, amplify, oscillate, filter, voltage divide, etc.) and, in the event of a malfunction, provide a recognizable symptom. The functional blocking, or the arrangement of one or more circuit elements into functional entities, is of major importance in the generation of the schematics. Normally, these blocks are blue shaded, but may be coded with dot-dash lines when permitted by contractual specification deviation. Every element must be contained in a functional entity block. Each block must be assigned an identifier code as described in MIL-M-24100. Similar circuits should be drawn in the same form to aid in technician/user recognition. In the strictest sense, a functional entity consists of the smallest number of circuit elements that perform a separately definable function (e.g., filter, oscillate, amplify). But there are other factors which must be considered when blocking, such as maintenance philosophy, technician usability, MDC generation, and the utilization of the functional entities on the functional diagrams. This gives rise to a less strict but far more usable definition. That is, a functional entity can be a grouping of interrelated circuit elements whose operation is best explained as a unit, and whose output(s) can be readily evaluated to verify desired operation. For example, consider the case of two transistor amplifiers in series, with capacitive feedback from the second to the first. Separate blocking of each amplifier and the feedback capacitor would be possible, but would make explanation of circuit interaction and desired output unwieldy. Presentation as a single functional entity provides a coherent block of information for the technician/user. Another important consideration in the definition of functional entities is the lines entering and leaving a blue box. Any line entering a blue (functional) box must act upon or be acted upon by the functional entity contained therein. The resultant output(s) should leave the box only once. Example A below illustrates some errors in blocking that can cause confusion on the major function diagram. Line X has no reason to go through the box at all. Line Y should branch before it enters the box, and line Z should branch after it leaves the box. An attempt to illustrate N-FL-3 on a major function diagram (example B) would result in a symbol with two inputs and four outputs, an obviously erroneous conclusion. Example C shows a correct method of blocking N-FL-3, and its corresponding function diagram representation. Notice that TP1 has been moved outside the blue box. This should be done whenever possible, as it provides a much clearer functional diagram picture of the point actually measured at TP1.



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Even with this simple example, the corrective interaction between the function diagram and the blocked schematic can be seen. Schematic blocking will at times have to be modified when generation of the functional diagram provides a more comprehensive picture of what is going on. Accurate preparation of the functional diagram requires line checked blocked schematics, but allowances should be made on the schematics for modifications to blocking. Recognizing possible problem areas and allocating room for corrective action are skills that will be developed only by actual preparation of blocked schematics and functional diagrams.

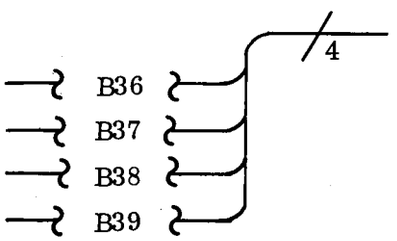
3.3.2.1.2 Functional flow. The left to right convention for functional signal flow was adopted because it is the way most people are conditioned to scan a written page. Inputs to a hardware enter the diagram on a left hardware boundary. Outputs leave the hardware on a right hardware boundary. Functional entities, in most cases, follow the same convention, with inputs on the left and outputs on the right. Functional entities should be arranged to maximize visibility of major signal functional flow. Major signal visibility is also enhanced by the use of a heavier line weight and appropriately larger signal arrowheads than other signal lines. In any decision concerning layout, major signal flow functionality takes precedence over other signal lines. This should not be misconstrued as license for haphazard arrangement of signal lines and functional entities not directly involved with the major signal path. All items should be arranged to optimize overall functional clarity, with a minimum of signal line dog-legs (bends) and crossed lines. In many cases, break-lines can be used to alleviate the confusion caused by many crossed lines. This is particularly true of the power break-lining techniques associated with the phantom power distribution functional entity designated C-PWR-1. All power and returns that enter the hardware immediately enter C-PWR-1. Power lines are broken inside the box and are assigned appropriate labels. Returns enter the box and are terminated in return symbols. Use of assigned symbols shall be consistent to illustrate the grounding tree structure. Power connections to the functional entities are indicated within their functional block by corresponding break-lines and triangle symbols. The following example illustrates some common practices that can be used in C-PWR-1.



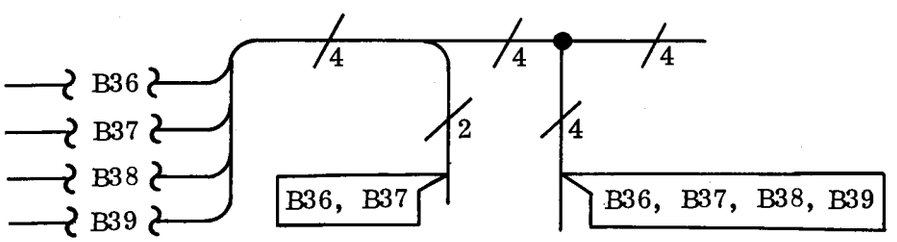
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Another use of the broken line is to minimize the crossing of lines having the same line weight. This technique differs from the distribution type break line in that the initiation of the break must be directly in line with the termination of the break. Some of the practices adopted for this use of the broken line are: when one line crosses two or more lines, break the single line; when a horizontal line crosses a vertical, break the vertical; breaks over long distances should be identified at each end; and minor lines crossing the heavy major lines need not be broken. Another practice used to unclutter drawings is that of bending hardware boundaries. Situations will arise where a signal enters the hardware, goes to one functional entity, and leaves the hardware. In conventional engineering drawings the practice has been to bend the output around and have it exit the hardware on the same boundary as it entered. In most cases, this does not meet functional signal flow requirements. Rather than having long lines running from one side of the page to the other, the drawing is arranged so that the functional entities involved in such a situation appear near the top or bottom edge of the hardware. The shape of the hardware boundary can then be distorted to allow the signal to come in a left boundary and/or go out a right boundary with a minimum line length (see figures 10 and 11). Another technique might be to divide the hardware into several parts and label them "part of" (P/O).

3.3.2.1.3 Composite lines. Of course, there will be situations when long lines are necessary to connect functional entities at either end of the drawing. These can be grouped into a composite line. Composite lines are used to group similar signal lines into a single line in order to avoid cluttered diagrams. Each signal line entering the composite line is identified, and curves as it enters the composite line. The number of unique lines is indicated with a slash and a number. Refer to the diagram below.



The composite line curves at every 90 degree corner in order to avoid confusion with single signal lines, which make right angles at corners. Distribution from the composite line is shown by means of a curved breakoff line (only a portion of the total signals in the composite line) or a straight line breakoff (all of the signals in the composite line).



Note that the number of signals leaving a composite line do not always add up to the number of signals entering the composite line. This is because some of the signals have multiple destinations. This is especially true in digital equipment, where the composite line technique is most commonly used.

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3.3.2.1.4 Parts identification. Parts comprising the circuitry making up functional entities should be fully identified. Reference designations, types, values, and any other identifying information not covered in the general notes should appear next to the circuit elements. This includes breakdown diode zener voltage levels, power transformer secondary voltages, and any other significant information that will aid the technician/user.

3.3.2.1.5 Analog with discrete components. Figure 9 is an example of a blocked schematic diagram for a series regulated power supply. It clearly illustrates that functional presentation of a circuit is the primary requirement, and may sometimes necessitate slight interpretation of standard FOMM blocking rules. This hardware was wired directly into the equipment, and had no identifiable connectors. To meet interconnection trackability requirements, identifiable connectors in another hardware were illustrated. In a few cases, connections to functional entities were made through the top or bottom of the functional block. For emitter, collector, control, and reference voltages this is sometimes allowable. Unlike most single component functional entities, Q4 was given the mnemonic code Q-SRG-2 to underline its function in the circuit. Notice that there is no C-PWR-1. This is often the case with power supply diagrams where no actual power distribution takes place. The input voltages to the hardware each have only one destination. The question might be raised that since returns R1 and R2 are actually the same return, tied together external to the hardware, why not tie them together internally and show them as one R? This is not permitted because it would cause confusion for the technician/user, as it would not indicate which circuit elements were connected to TB1-7 and which to TB1-3.

3.3.2.1.5.1 The practices illustrated by the schematic are fairly standard. Circuits are shown in their textbook fashion. For example, series regulators are shown in common base configuration, difference amplifiers are shown with emitters facing, and voltage regulators are shown in shunt to ground. Another of the general practices used in making clear diagrams is the orientation of the symbols. Resistors that perform a current limiting function, like R9 of Q-SRG-1, are placed horizontally, while those that form part of a divider, load, etc., are oriented vertically. Capacitors that are used for coupling are shown horizontally, while those used for filtering are shown vertically. Diodes which perform a rectifying function, like those in X-RCT-1 and X-RCT-2, are horizontal, while CR7 of Q-DFA-1, which performs a clamping function, is shown vertically. Power transformer secondary voltages and breakdown diode zener level voltages are indicated. Major functional flow, as well as secondary, reference, and feedback paths are so noted by the appropriate line weight and arrowheads.

3.3.2.1.6 Analog with integrated circuits. Figure 10 is an example of a blocked schematic for this type of hardware. It clearly illustrates the practice of drawing similar circuits in the same form. Screwdriver adjustment symbols are cut back to the hardware level on which they are available. Gate, source and drain connections for the field effect transistors are indicated. Other conventions for the presentation of discrete components were discussed previously. The major difference on this diagram is the inclusion of integrated circuits. Contained within each symbol is its reference designator and the label for its inverting (-) and non-inverting (+) input. Pin numbers are called out external to the symbol. Circuit elements and connections made to the chip which are not in series with the signal (e.g., roll-off resistors, capacitors) should be shown at the top and bottom of the symbol. Positive power connections should be indicated at the top of the symbol, negative at the bottom. When multiple-use chips are involved in the performance of different functions, their integrated circuit symbol should be repeated. A letter should be added to the reference designator (e.g., AR5-A, AR5-B) to indicate this separation. Power connections should be made to only one of the repeated symbols.

3.3.2.1.7 Digital with integrated circuits. Reference the digital schematic example in MIL-M-24100. Immediately noticeable is the lack of a major signal flow line. All the signals are approximately equal in importance, and it would be impractical to make them all heavy lines. This situation is quite common with digital equipment. A major problem in generating the blocked schematic for this type of hardware is the large number of interconnecting lines. One technique that has proven useful in reducing drawing clutter concerns power connections to the integrated circuits. When all ICs have power connected to the same pin number, a general note so indicating will suffice. Often the hardware contains several different IC types, with power connected to different pins on each type. In that case flag notes can be used. Flags are situated in C-PWR-1 next to the power and return involved. The connections are then listed in the flag note box (see figure 11).

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Another technique, found helpful in reducing crossed lines within functional entities, involves the use of distribution-type break lining (see the KEEL REFERENCE DECODER blocks at the right of the example in MIL-M-24100). Signals can be broken inside the blue box and assigned an identification letter. Intended connection points are so noted by a corresponding letter and break line.

Note that, on digital schematics, the first level blue boxes can have functional names rather than mnemonic codes.

Information concerning symbology and operation for each type of integrated circuit used must appear on the IC data page of the manual. Symbology must be consistent with current military and industry standards.

3.3.2.1.8 Analog/digital hybrid (see figure 11). This is another example of all signal lines being relatively equal in importance, so no heavy major signal flow line is used. The conventions concerning schematics for each type of hardware cover the associated portion of the schematic to which they are applicable. For example, power connections to the digital ICs are handled by flag notes, while analog IC op-amps have power break lines. Several other practices are also illustrated. Resistor-capacitor charge path should be shown vertically, similar to the one at the emitter of unijunction transistor Q2 in Q-OSC-1. Also note that the bases are identified. Unused or spare components should be accounted for by notation in a "spare element" box.

3.3.2.1.9 Electromechanical schematic. The presence of motors and gear trains create a need for a special set of techniques (see figure 12). The problem is to illustrate the electrical and the mechanical portion of the diagram while indicating the relationship between the two. In the example this is accomplished by showing the motor (B1), and the potentiometer (R1), both symbolically and pictorially. The symbolic representations are tied to their pictorial representations by dashed lines. This convention is used to indicate a mechanical or magnetic linkage. Note that the mechanical action resulting from the major signal drive is illustrated as a major line. The techniques for representing gear trains are somewhat varied. The example illustrates some of the basic conventions, such as indicating gear meshes and number of teeth. The depth of detail will depend on the source data available and the maintenance philosophy. If the technician is expected to repair the gear train, the information should be quite detailed. Direction of motor rotation should be indicated. If, as in the case of figure 12, the motor can rotate in either direction, a direction of rotation could have been assumed, and the resultant gear rotation indicated. The important thing to stress is the functional operation of the mechanical assembly. In figure 12, the technique used shows the North-South gear train moving a gear which is adjacent to the East-West assembly and is labeled NORTH-SOUTH. Similarly, the East-West assembly is depicted as moving the sliding metal screen in the EAST or WEST direction.

3.3.2.2 Schematic text. The schematic text describes what the hardware is, what it does, and how it is done. The information is presented as a combination of two formats (see figure 13). There is a general explanation of hardware function, geared to the overall operation of equipment. This is presented in paragraph form, and precedes the keyed text. The keyed text contains a detailed description of the operation of each functional entity on the diagram.

3.3.2.2.1 Text numbering. The order in which numbers are assigned to functional entities is an important consideration. Two general practices have been adopted to provide a logical flow of information and avoid confusion. First, text numbers are assigned following the functional signal flow. They usually start at the upper left of a drawing and work toward the right, along a particular signal path. Then, when a number of signal paths lead to one functional entity, operation of each of the entities along the convergent signal paths is explained before the one for which they are inputs. Refer to the functional entities numbered 5, 6, 7, and 8 on figure 13.

3.3.2.2.2 Depth and level of writing. MIL-M-24100 specifies that the troubleshooting/repair volume be written for a high-school graduate who has previously completed military courses qualifying him as a technician. This, in most cases, makes unnecessary the repetitious explanation of common circuit items normally included in a functional entity (e.g., transistor biasing resistors, interstage coupling capacitors). Emphasis should be placed on circuit operation as a functional entity, with detailed explanations of individual components as required to enhance technical clarity (note the detailed information concerning CR4 in text circle 5, figure 13). As for the text itself, keep sentence structure simple. Studies have shown that reader comprehension drops off rapidly when sentences go beyond 17 words. Conversely, don't replace two or three common words with one uncommon word just to keep sentence word count down. The object of the text is user/technician comprehension, and a dictionary might not be readily available.

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3.3.2.3 Hardware maintenance dependency chart. Hardware level MDC's are prepared when dictated by the equipment packaging concept, the maintenance philosophy, and the level of coverage required by the contract. If the equipment is completely modularized and the maintenance concept is to replace and scrap faulty "plug-in" boards, then hardware MDC's are not required. Even when there is a requirement for repair information, there may still be portions of the equipment which do not warrant an individual MDC. An example of an assembly not requiring a hardware-level MDC is a control panel in which isolation to a faulty component is fully treated on the major function MDC. Another example is a heat sink mount for a transistor which, from a functional view, is a portion of another assembly. In some cases the hardware MDC may be supplemented or replaced by other types of maintenance data such as instructions for continuity checks, front-to-back resistance ratios, backlash tests, etc.

3.3.2.3.1 Classification of hardware MDC's. All portions of the hardware MDC can be influenced by whether the fault isolation is to be performed with the assembly mounted in the equipment, or at a test bench. If the MDC is written for use at the equipment, then the procedures, measurement points, and signal specifications closely follow the major function information. In fact, if it were not for the desire to limit the complexity of the major function MDC, there would be no need for a hardware MDC referenced to equipment operation. When troubleshooting is to be performed at a test bench, then the hardware MDC reflects the tests specified by design engineering and may, from the standpoint of the type of tests, method of connection, and test results, have little which is recognizable with operation in a system. For the writer who makes a hardware MDC, it is normally easier to prepare one which is to be used at a test bench, since the guiding hand of engineering is provided in the form of specifications and procedures. Preparation of the MDC referenced to system operation requires more insight into the equipment to ensure that the troubleshooting procedures completely exercise the functional entities within the hardware, and that the results recorded in the signal specifications allow an unambiguous determination of proper operation.

3.3.2.3.2 Technique for preparing hardware MDC. Figure 14 is an example of a hardware MDC. This is the MDC for the schematic shown on figure 9. Note that the instructions reference switches which are not part of the assembly, and that the heading identifies measurement points on a different assembly than that under test. This information indicates that the MDC is used at the equipment rather than at a test station. The sparsity of the preliminary instructions occurs when the method of turning the equipment on is well understood by the MDC user, and when the full instructions are documented elsewhere. The first step in the instructions specify particular system controls which must be set to ensure that power is supplied and that a given load is connected to the circuit. In this simple example, only one step is required to prove that the circuits are operating properly. The second step is an example of the information which should be provided to aid fault isolation within a feedback loop. Use of this step by the troubleshooter is only required when the results of step one are unsatisfactory. If the hardware MDC required several steps, then step 2 would be shown as a sub (checkout) step in order to emphasize that the procedures given should only be performed after determination of a failure.

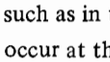
3.3.2.3.2.1 Rules for substeps must be explained on the "How to Use MDC's" page - i.e., the switch settings in a substep represent only changes from a major step, not an immediately prior substep. If a sub-substep must be used, the switch settings must show only the changes from the immediately prior substep.

3.3.2.3.2.2 The layout of the heading and the body (dependency lines) of the chart is where the writer demonstrates his skill in clearly presenting the troubleshooting path. Despite all the rules and examples given in the specification, there is still a certain amount of personal preference permitted, and consequently it is not likely that two writers, working independently, would produce identical MDC's. Following is a discussion of how the MDC of figure 14 could have been constructed by a different writer - or by the same writer if he spent more (or less) time in preparation.

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3.3.2.3.2.3 For an MDC with this simplicity, the only rules which clearly apply are that inputs are to be shown in the first (left) columns and outputs are to be shown in the last (right) columns. Additionally, if dependency exists from circuit to circuit, then the slope of the overall structure should be diagonally from upper left to lower right. There are no rules stating in which order the inputs are to be shown. One writer may decide that signal specification number one should be for the largest voltage and that other inputs be assigned specification numbers by descending voltage values. A second writer may make the opposite decision and a third writer may use no system at all. In the body of the chart, one writer may quite legitimately decide that because signal specification 15 is not dependent upon 14, then dependency line three should really be shown first. Another writer may decide that the development of signal specifications 14 and 15 are completely parallel to most of the circuitry shown, and therefore the columns showing the development of these two outputs (X-VR-1, X-VR-2, signal specification 4, and Q-SRG-1, C7) should all be moved to the far right. Because there are many ways of constructing the MDC for a given circuit, and because the usefulness of the information is highly dependent on the visibility provided by the dependency structure, it is suggested that the method for construction be as follows: (a) Enter heading information measurement points and functional entities) in the same order as indicated by the signal flow path on the schematic but leave every second or third column blank. The blank columns may eventually be used for functional groupings overlooked during the initial compilation or to ease the burden of rearrangement. (b) After the heading information is entered, construct the MDC and then rearrange columns, add or delete measurement points, as required to satisfy specification requirements for emphasizing serial/parallel relationships, convergence, divergence, etc. (c) Check the MDC to ensure that performing the steps will fully check all the circuits, including failure detection circuits, and then leave the MDC in its rough draft form until it is fully checked against operating equipment. The reason that the MDC should be left in rough draft form is that quite often the writer will not know if the measurement points he has selected are accessible or if the test results appear to have the validity he expects. Also, in some cases, while performing the checks, the writer may have his first opportunity to receive practical data from those most knowledgeable in troubleshooting the equipment. (d) After the MDC has been corrected and inked, it should be completely validated against the equipment. If this is not possible, then validation should be by a line-for-line check against the functionalized schematics which, in turn, have been validated against the engineering drawings.

3.3.2.3.2.4 The signal specification table is prepared after the body of the MDC has been completed. All of the signal specifications should be measured on equipment known to be operating to design requirements using the test equipment specified. Pay particular attention to the accessibility of measurement points. All measurements should be compared with a theoretical analysis of the circuitry to ensure proper timing, rise times, power levels, tolerances, and synchronization to other parts of the equipment.

3.3.2.3.2.5 Many MDC's are much more complicated than that shown in figure 14, and some may require the use of special symbology not shown in the example. The sanctioned special symbology is shown in the specification and, if the basic rules for a MDC are understood, then the use of some special symbology should be self-evident. The exception is the use of contracted events and, by not providing examples, the specification implies that it is preferable not to use these symbols. Contracted events can be used to reduce the size and complexity of the MDC, but at the expense of increasing the amount of detail in the signal specification table. Two special symbols are required when the contracted event technique is used. They are the contracted event (availability) box and the combination dependency marker. A contracted event box contains the same information as a regular event box plus a number(s) such as in the following example  . This indicates that for the dependency structure shown, outputs should occur at three different measurement points and that these outputs are identified in the signal specification table. The signal specification table must then contain a tabular listing of the signal specification identification (e.g., 4.1, 4.2, 4.3), the applicable measurement points, and the characteristics of the signal at each measuring point. The combination dependency marker is used in conjunction with the contracted event box and may appear as follows  The number within the triangle indicates that there must be at least two of the signals present out of the total number shown in the contracted event box,and the P indicates that it must be a particular two. Identification of which particular two signals are required is included in the signal specification table as part of the data concerning the available signal resulting from the operation of circuits using the particular signals. From the above discussion it may be concluded that, even if the writer understands and uses contracted events properly, there is still a question of whether he can communicate all his knowledge to the MDC user. However, contracted events do have a place,

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such as in MDC's for three phase power circuits. A second case in which contracted events are often used is in MDC's for logic circuits using parallel bits of information, however, with parallel bits, care must be exercised. If contraction is carried too far, every dependency line may look the same, and the MDC data becomes identical to that which can be provided in words or in a truth table, and the method of presentation makes the information almost unusable.

3.3.2.3.3 Hardware maintenance data. It may be noted that the specification does not define what constitutes corrective maintenance data. There are two reasons for this: first, the justification for the T/R Volume is that any or all of the included information may be required to perform corrective maintenance. Therefore, everything in the volume is maintenance data. Secondly, the definition for corrective maintenance data is normally defined as the material to be contained in a particular chapter or section of the technical manual. The latter definition is not used because the T/R Volume is divided into data packages (rather than by chapters), and the information within a package may serve more than one purpose. Even though there is not clear delineation of what constitutes maintenance data, it is still desirable to establish some definition in order to ensure the listing of all information in the functional index, and to establish rules for the sequencing of information within a data package. The adopted definition is that (by name association) the MDC and its signal specifications is maintenance data, and all information in a package not required (or substituted for) the basic contents (diagram, text, parts data) is also maintenance data.

3.3.2.3.3.1 Figure 15 is an example of a page containing maintenance data. In this case the associated hardware consists of motor driven potentiometers for which a MDC is not included,and therefore the information shown is the only maintenance data given. This information is on the same page as the parts data so that easy reference is made to the picture and mechanical schematics. The maintenance data given is typical and may be expanded or contracted as necessary. The procedures for any maintenance data should be a set of step-by-step instructions written in third person, active voice, imperative mood. All necessary information for the performance of the steps should precede the procedures. This preliminary information should list test equipment, power requirements, and, if it is not obvious, discuss the rationale for the procedures. Cautions, warnings, and notes may be summarized and placed just before the first step of procedures. If there is any possible question of applicability, or if the procedures are lengthy, the warnings, cautions, and notes must also appear in the body of the procedures. It is suggested that validation of the maintenance data be performed by a novice with the writer as a witness. If it is necessary for the writer to give any verbal instructions during validation, then obviously the procedures require a rewrite or rearrangement.

3.3.2.4 Alignment. Normally, alignment procedures included as part of the maintenance data are edited versions of engineering documents. The problem is that there may be several engineering documents covering alignment at the hardware level, equipment level, etc. It may be the task of the writer to determine which procedures to use and where they are to appear in the volume. The writer's decision should be based on the established maintenance philosophy for the equipment,and must also include those "factory adjustments" which could be used in a last desperate attempt to restore equipment operation. The latest maintenance philosophy for modularized equipments dictates that,wherever possible, replacement of "plug-in" assemblies be accomplished without affecting system alignment. Unless the writer receives contrary information, he should adopt this remove/replace philosophy. This implies that all the alignment information is included within the hardware maintenance data except for those adjustments which are truly interacting. Adoption of such an alignment requirement may complicate the work of the writer,for now he must provide instructions necessary to precisely set a circuit transfer function. This can require drawings of special test set-ups, instructions for disconnecting and reconfiguring the connections to some devices, and descriptions of the pertinent features of waveforms, etc. When "factory select" or "factory adjust" is indicated, the criteria for selection or adjustment must be included. The method of presenting alignment instructions should, in general, follow the same format as specified for other maintenance data.

3.3.2.5 Parts data. Parts data will always be included in the hardware data package for each item designated repairable. Two basic items constitute the data: a parts location illustration, and a parts data table. A source data block is also required. Additional data items, such as usage table and auxiliary views, should be included as necessary to provide complete parts information.

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3.3.2.5.1 Parts location diagram. Complete parts location information must be included for each parts data table. For a circuit board or chassis, the preferred presentation is line art illustration, but a good quality half-tone may be used (see figure 15). Two parts location methods may be used: a locating grid, or leader lines. The locating grid can be either blue overlay or a zone indicating border. In figure 15, the zone indicating border type was used because the blue overlay would have diffused the clarity of the half-tone. Leader line call-outs are usually associated with hardwares having few parts, or mechanical assembly exploded views.

3.3.2.5.2 Parts data table. The contents of the parts data table will depend on contractual requirements. If a separate parts list is specified by contract, the parts data table need contain only the part reference designator, its generic name (e.g., resistor, capacitor), and its locating coordinates. When parts data appears only in the hardware packages, the data table must also include parts ordering data. The specific information to be included is covered in MIL-M-24100. Items with mechanical part (MP) designators are usually assigned these designations by the person preparing the parts data. Common attaching hardware (e.g., screw, nuts, washers) need not be included in the parts list unless referenced in an assembly/disassembly procedure, or made of materials of specialized characteristics. There are two columns in the parts data table appearing in figure 15 not discussed in MIL-M-24100. These columns are used to specify actions to be performed at certain maintenance levels, and are usually associated with parts lists prepared for the Navy.

3.3.2.5.3 Additional data items. Two additional data items, the usage table and the assembly auxiliary views, are included in figure 15. These are not necessarily required, but exemplify the kind of additional information that should be included to aid the user.

3.3.2.6 Wiring diagram. Wiring diagrams or wire run lists are required for each chassis, backplane, and front panel assembly. For backplanes, a double-entered wire run list can be substituted. The list must present the wires in alphanumeric order. Figure 16 illustrates a typical wiring diagram. It is usually copied from engineering source data, including explanations of any special techniques used in the presentation. The schematic diagrams are electrically correct and functionally oriented. The wiring diagram is more hardware oriented, reflecting the actual point to point wiring of the equipment. Wiring diagrams should identify components, terminals, wire number and color coding. Shielding and shield grounding must be indicated.

3.3.3 Major function data package. The major function data package consists of a functional block diagram, supporting text, and an MDC. There shall be one data package for each major function of the equipment, as defined on the overall block diagram. Some of the tradeoff considerations for dividing the equipment into major functions are:

(a) The inherent ability to isolate a potential fault, detected at the equipment level, to a major function.
(b) The complexity and amount of circuitry required to mechanize the major function.
(c) The final page size of the manual.
(d) The inherent ability to isolate a potential fault, from the major function, to a given hardware assembly.
(e) The mean-time-to-restore requirements.

3.3.3.1 Major function diagram. There are four types of major function diagrams. The basic purpose of the diagrams is the same for all four; describe a major function of the equipment. The four types are:

(a) Power distribution diagrams
(b) Major function diagram - analog
(c) Major function diagram - digital
(d) Ground/return diagram

Although the four types of major function diagrams follow the same basic rules, emphasis is shifted depending on the mechanization used in the equipment. Each of the four types is discussed in 3.3.3.1.1 through 3.3.3.1.4. The following is a list of rules that should be applied to major function diagrams.

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(a) All functional entities on all schematics must be represented by a symbol on one of the major functions.
(b) Each functional entity symbol on the major function shall contain the same mnemonic code used on the blocked schematic.
(c) Functional entities shall be grouped into subfunctions.
(d) Signal lines shall be identified. Identification shall be consistent with corresponding signals on the overall diagram and the blocked schematics.
(e) Adjustments, controls, and test points must be shown on the proper hardware level of containment.
(f) Hardware reference designation must appear in the upper left corner of each hardware boundary.
(g) Flag note block must contain complete information concerning each flag note used on the drawing.
(h) Signal flow must be from left to right and top to bottom.
(i) Front panel controls must be cut back to a white background and given a heavy shaded border. A dashed line shall be used to "connect" the control to its block diagram symbol.
(j) Shielding essential to circuit operation shall be shown.

3.3.3.1.1 Power distribution diagrams. Power distribution diagrams are used to show the distribution of power, from the point where prime power enters the equipment, through all connectors, fuses, switches, etc., to the lowest level of replaceable subassemblies. In simple equipments, only one power distribution diagram is required. When more than one power distribution diagram is required, the first division should be between a.c. and d.c. power. Further divisions should be by frequency voltage level and sections of the equipment (i.e., Unit 1 +28V Power Distribution Diagram).

3.3.3.1.1.1 Alternating current (a.c.) power distribution diagram. This diagram is listed first, because it should be prepared first. Figure 17 shows a relatively simple a.c. power distribution diagram. As mentioned in the section on horizontal trackability, the breakline/interface rules must be defined and followed. In this example, the a.c. voltage and mechanical linkages to a.c. and direct current (d.c.) power distributions are interfaced. The d.c. power coming into the a.c. distribution is breaklined. The functional inputs are interfaced. The example shows the detail of the cabinet interlock switch. Such details should be included when operation is not obvious. Switches on block diagrams should be presented in a way that demonstrates their operation. In the example, S1 serves as an illustration. When the OFF/ON switch is rotated "down", continuity is made between pins 5 and 7 of the switch and the two "down" contacts, pins 6 and 8. Relays are always shown such that the contact closest to the solenoid makes connection when the relay is energized. The relay solenoid is given a name indicating what it does when energized. The four relays toward the left side of figure 17 remove power to four lamps when energized, and are named accordingly. The same left-to-right flow techniques that apply to schematics apply here. In a power distribution there are usually many parallel paths, which makes major flow somewhat subjective. In many cases, as in the example, no major flow is indicated.

3.3.3.1.1.2 D.C. Power distribution diagram. The d.c. power distribution diagram is quite similar to the a.c. in that both stress distribution. Refer to the example in MIL-M-24100. All assemblies that receive the power from the distribution should be shown, with the power terminating in a blue box. The complete grounding path should also be shown. If there are any components inside the terminating C-PWR-1, they are shown in block diagram form. The example in the middle of the page illustrates this convention. Similar circuits should be shown in a similar way. In the example there are two 25V power supplies which are identical except for external connection. They are drawn the same, which makes it necessary to explain only one in detail. Note the use of special arrowheads. They are quite common in d.c. power distribution diagrams. A hollow arrowhead (output of Q-DET-1 at the bottom of both power supplies) indicates a fault signal, in this case a shutoff voltage. Feedback arrowheads are used to identify the feedback path of the series regulator. Indicator signals, such as those going to the two meters,are given their characteristic arrowheads.

3.3.3.1.2 Major function diagram-analog. Figure 18 illustrates a major function for an equipment of moderate complexity. The entire functional representation is combined on one diagram. In such a case, this would also serve as the overall diagram. Though functional symbols are normally used, schematic representations can be repeated on the block diagram if this results in a more understandable diagram. In figure 18, the CRT, chart motor, and recorder keying switch are illustrated schematically for this reason. On the function diagram, line art symbols represent the

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functional entities shown on the schematic diagrams. These symbols are grouped again in blue boxes according to their interrelated operation (e.g., DEFLECTION GENERATOR, RECEIVER, CRT DISPLAY). This enables the writer to provide a single, coherent explanation of operation without unnecessary cross referencing to other blue boxes.

3.3.3.1.2.1 Major function diagram - electromechanical. An electromechanical major function diagram is an extension of the analog major function diagram. In addition to following the same techniques for the electronic portions of the major function, the mechanical portions must be shown as pictorially as possible. The objective is to show the interrelationships between the electronic and mechanical portions, and to provide the user with a clear picture of the mechanical output. Figure 19 illustrates some of the techniques that should be used on an electromechanical major function diagram. Note the heavy dashed lines used to indicate the continuation of major signal flow after its conversion from electrical energy to mechanical energy. Gears are represented by diagonally divided boxes, with gear ratios indicated in the appropriate division.

3.3.3.1.3 Major function diagram-digital. Refer to the example in MIL-M-24100. Here again, line art boxes representing schematic functional entities are grouped and given names according to their interrelated actions. The same techniques used on a digital schematic (e.g., composite lines, break lines) are used on a digital function to help alleviate problems caused by many crossed and paralleled lines. As with digital schematics, major signal flow is sometimes difficult to define in a digital major function diagram. In this example, the major signal is the ECHO, which is gated by the digital function. Many of the timing signals are also important. However, with digital circuits, to make one such signal major would necessitate making many signals major, and result in undermining the concept of major signal flow. MIL-M-24100 states that there can be no overlapping blue and grey shades. Thus, each blue shade must be entirely within one grey shade, and any grey shade enclosed in a blue must be entirely within it. The interface boxes (blue) at the lower right and lower left of the example have one or more grey shades inside them. This technique was used to conserve space, as 12 interface boxes would have to be shown instead of 7, if each blue interface box were enclosed in a grey hardware. The rule of overlapping blues and greys also restricts the functional grouping of the block diagram. In the example, the keel reference switches (S1 and S2, lower center of the diagram) are really associated with the keel reference counter. However, because they are in different hardwares, they must be blocked separately. The switches and lamps of this example were shown schematically to clarify their function on the block diagram.

3.3.3.1.4 Ground/return diagrams. Ground/return diagrams are required only when the complexity of the grounding scheme is such that it cannot be shown on the power distribution diagrams. When a ground/return diagram is prepared, it must show the entire grounding scheme, not just a portion of it.

3.3.3.2 Major function text. Major function text explains what the major function does, and how the functional entity groupings on the major function diagram contribute to the operation of the function. The structure of the text is similar to the hardware level text; that is, it begins with an opening paragraph or paragraphs explaining what the entire function does, and is followed by keyed text paragraphs which explain what each blue box on the major function diagram does. Figure 20 illustrates a portion of the text written for figure 18. The opening paragraph(s) should relate the operation of the major function to the overall equipment operation. There will be some overlap between a keyed text paragraph of the overall function and the opening paragraph of the major function text. There is also some overlap between keyed text for the major function and the opening paragraph of the related hardware schematic text. The reason for this overlap is rooted in the vertical trackability feature of the FOMM diagrams. Because they overlap, their associated text will overlap also.

3.3.3.3 Major function MDC. At the major function level, the maintenance dependency chart is the primary troubleshooting vehicle. Since the MDC is closely related to the corresponding major function (and developed from the major function diagram), it is imperative that the diagram be accurate before attempting to construct the MDC. This requires a complete line check of the diagram. MIL-M-24100 requires that each hardware assembly containing functional entities be represented in the heading of the MDC. Each input to the major function and each output from all assemblies must be entered in the heading and assigned a signal specification number. The trick in constructing a major function MDC is arranging the headings so that the dependency structure in the body of the MDC flows from left to right. Refer to the paragraphs concerning hardware MDC's for additional rules and guidelines for MDC development. An example of a major function MDC appears in MIL-M-24100.

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3.3.3.3.1 Fault logic diagrams. Another method of presenting fault isolation information is the fault logic diagram (figure 21). This may be used for equipment that is predominantly digital in nature. However, it is imperative that the fault logic diagram provide the same depth of coverage as would be realized if an MDC were used. That is, all circuits utilized must be checked in every mode of operation. The fault logic diagram must be accessible from the overall level diagram, and must direct the technician/user to the appropriate hardware level troubleshooting data.

3.3.4 Overall function information. The overall data package will normally be the manual user's initial entry point into the troubleshooting/repair information contained in volume 2. The basic presentation of information consists of an overall diagram, text, and an MDC. Depending on the type and complexity of the equipment, options include a fault logic diagram (replacing the MDC), and intermediate diagrams with fault isolation instructions as required.

3.3.4.1 Overall diagrams. The overall diagram illustrates the interrelations between the major functions into which equipment has been divided (see figure 22). Preparation of the diagram follows the same functional conventions as those for other diagrams; all electrical and mechanical connections between functional blocks must be illustrated; left to right signal flow; identification of signals by line weight, arrowheads, and signal flags; and minimization of crossed lines using composite and broken lines. The way the functional blocks are arranged on the diagram, and the order in which text numbers are assigned, determines the order in which corresponding major (and intermediate) function diagrams will appear in the volume. Front panel nomenclature, indicators, and control illustrations used on each major function diagram will be repeated on the overall diagram, in the blue box representing that major function. Intermediate diagram representations on the overall diagram will not contain front panel illustrations.

3.3.4.1.1 Overall with intermediates. Intermediate diagrams are generated when the overall diagram exceeds its page size limitation. Other than front panel illustrations, there is no difference between the representation of an intermediate on the overall diagram and the representation of a major function. Figure 23 is an example of an overall diagram with intermediate functions.

3.3.4.1.2 Intermediate diagrams. An intermediate diagram is a convenient grouping of interrelated major functions which, because of page size limitations, cannot be individually represented on the overall diagram. Preparation of an intermediate diagram is similar to an overall diagram. Refer to the example in MIL-M-24100. The same functional flow and crossed line minimization techniques are used as on other functional drawings. Major function front panel illustrations and nomenclature included on an intermediate diagram are not repeated on the overall diagram.

3.3.4.2 Overall text. Overall text is presented in the standard FOMM format: opening paragraphs providing an overview, followed by text keyed to the functional blocks on the diagram. Figure 24 contains the text for the overall diagrams shown on figure 22. The opening paragraphs in this case are particularly sensitive in that they are the user's introduction to the FOMM manual functionalization of the equipment. The information must bridge the gap between the theory of operation in the support volume, and the functional mechanization of that theory by the equipment, as described in the T/R Volume.

3.3.4.3 Overall MDC. Refer to the example in MIL-M-24100. The overall MDC provides fault isolation information to a major function. The heading of the chart is divided into major functions rather than hardwares. The instructions in the procedure column have a one-to-one correlation with the steps in the turn-on/check-out chart. The techniques for preparation of the body of the chart are the same as for any other MDC. There are, however, some special techniques which can be used with extremely complicated equipments. If intermediate diagrams have been generated, and the overall MDC is quite involved, it may be beneficial to generate intermediate MDC's to accompany the diagrams, thus alleviating some of the clutter from the overall MDC.

3.3.5 Additional data items. In previous discussions, emphasis has been placed on the similarity of content for data packages at each hierarchical level. That is, a data package at any level consists of a diagram, text, MDC with signal specifications, and other maintenance information. At the overall information level, there are additional items which have not been included as part of the upper hierarchy data package because there are no corresponding items in the lower level data packages. These items are called equipment introduction material and include the family tree diagram, interconnecting cabling/piping diagram, overall alignment instructions, and the turn-on/check-out chart.

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3.3.5.1 Family tree diagram. The tree chart presents a breakdown of the documented equipment by assemblies, with each box on the diagram representing an identifiable hardware (see figure 25). The information contained within a box is equivalent to nameplate data, or that contained in the title box of the engineering assembly drawings, and should include the following: (1) Reference designation or assigned contracting agency nomenclature; (2) common name; (3) manufacturer's part number or national stock number; (4) additional data which, in the opinion of the writer, will aid the manual user (e.g., equipment difference data). Vertical arrangement of the boxes should emphasize hardware subordination as determined by analysis of engineering parts lists and "next assembly" drawings. Horizontal arrangement should make the best use of the space available. As shown in figure 25, equipment redundancy should be shown rather than use "same as" statements. The depth of coverage, i.e., how detailed the hardware subordination, should be compatible with the level of schematic and parts list coverage in the manual. It is suggested that the first cut of the tree diagram be the first task in manual preparation. Then the chart can be used as a planning document for task assignments, recording of engineering drawing changes, etc.

3.3.5.2 Interconnecting cabling/piping diagram. This diagram depicts the connections between units of the equipment. Jacks, plugs, and junction boxes should be identified by their assigned designators. External connections to the equipment should be indicated as in the example in MIL-M-24100. This same diagram may be used in the installation section of the support volume. A list of cables is also required. It should identify all interconnecting cables, their type, number of active conductors, and the designations and types of end and mating connectors. Formatting standards are flexible for this item. Engineering source data may be reproduced if it meets content and legibility requirements.

3.3.5.3 Alignment. The alignment procedures that appear in a manual are usually the result of careful editing of engineering source data. These instructions are extremely important. Each procedure should be validated completely, with nothing left to chance. Required test equipment should be listed preceding the instructions. Adjustments should be grouped by major function. Start with the least dependent adjustments, specifying all required switch positions. Critical adjustments or dangerous situations must be preceded by an appropriate note, caution, or warning. One effective method of validation is to have someone who is unfamiliar with the equipment perform the alignment. If additional verbal instructions are needed, the procedures need improvement.

3.3.5.4 Turn-on/check-out chart. This chart should be a step-by-step listing of operator/technician actions to turn on the equipment, and what the resultant front-panel indications will be. It is the first step of the troubleshooting procedures. The chart is identical to the Overall MDC, except the body and headings have been edited to delete the functional entity symbols, availability symbols, and their corresponding headings. This leaves only the procedural steps and sensory indications with their appropriate headings. This enables the technician to move directly from the turn-on procedures to the appropriate area of the Overall MDC when he observes a faulty indication. The chart may also serve as an operational readiness verification in the support volume.

3.3.6 Volume 2 front matter. Paragraph 4.2 of the writer's guide describes the general front matter requirements for each volume of a manual. There are, however, three items that are strictly associated with the T/R volume. These contain detailed information concerning volume 2 symbology and data usage, and are described in 3.3.6.1 through 3.3.6.3.

3.3.6.1 Description of symbols, codes, abbreviations, and shading. This information enables the technician/user to properly interpret the information contained in the T/R volume. MIL-M-24100 details the required data items, and includes a figure illustrating many examples of the symbols, codes, abbreviations, etc., and their definitions. Other codes and symbols may be generated as required. Tables should include only those items used in the manual being prepared.

3.3.6.2 Foreword. The foreword to every FOMM manual should contain manual use instructions. These instructions should explain the organization of the manual, and the procedures to be followed in tracking a potential fault through the various data packages. These instructions are normally summarized in the support volume, and explained in detail in the T/R volume. When the T/R volume is divided into parts, the manual use instructions should be repeated in each part.

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3.3.6.3 Integrated circuit data. Since symbols representing integrated circuits are used on the blocked schematics, detailed information describing their operation must be provided for reference in the front matter of volume 2. The figure in MIL-M-24100 gives examples of applicable data, but it is not intended to be limiting. Any and all information that will aid in understanding integrated circuit operation should be included. When a large amount of IC data is required, it is suggested that the writer obtain the approval of the procuring activity to move the IC data to an appendix.

3.3.6.4 Explanation of MDC usage. The importance of this information cannot be over-stressed. The trouble-shooting effectiveness of the manual depends on its comprehension. An explanation of MDC usage is included in the front matter for every volume and part in which MDC's are used. The explanatory text used on the sample in MIL-M-24100 can be modified and used in most manuals, but the example diagram and MDC must come from the manual being prepared.

4. VOLUME 1 - SUPPORT VOLUME

4.1 Scope and content. The support volume provides information to set up, operate, and evaluate the operation of the equipment. This volume also contains parts information when a separate parts list is required by contract. Installation data is included, unless specifically excluded by the contract. Most of the information in the support volume is derived from engineering documentation and the procuring activity. The major tasks in producing the volume are arranging format, and ensuring that the data included is accurate and complete.

4.2 Front matter. This information identifies the manual and its contents, and describes how to locate and use the data provided. FOMM use instructions are generated strictly in-house (e.g., description of symbols, etc., explanation of MDC usage) and are similar from manual to manual. Items concerning information location (e.g., table of contents, functional index) cannot be completed until the page count and arrangement of the entire manual are firm.

4.2.1 Cover (see examples in MIL-M-24100). Cover information on each volume and separately bound part of a manual is identical, except for volume identification and part number. Manuals with 16 numbered pages or less (4 sheets, folded and saddle stitched) do not have covers. They are contained in clear plastic bags, using the title page information for manual identification.

There is usually only one publication number, as furnished by the procuring activity, shown on the cover. If the manual is to be used by more than one service, however, the procuring service's number is listed first, with the other publication number(s) immediately below it. Each service's name will appear before its number. If the manual replaces an existing manual, a supersedure notice, including the publication number(s) and date of the old manual, is placed beneath the publication number of the new manual.

Back covers are blank except for volumes (or parts) containing classified data. When classified data is involved, security information on the back matches in type size and style that which appears on the front.

4.2.2 Title page. The title page has the same information as the cover, with the addition of a distribution statement (transmittal notice) and publication date.

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4.2.3 Warning page. This page is not required in certain Navy manuals, but may be included in anticipation of multiple service use. It consists of the more vital warnings appearing in the manual. Some examples of the types of information are shown below.

WARNING

RADIATION HAZARD

This equipment contains the following radioactive tubes:
(List applicable tubes and equipment locations)
Radiation may be present at unsealed or broken waveguide elements.

Sample A.

WARNING

HIGH VOLTAGE

is used in the operation of this equipment
DEATH ON CONTACT
may result if personnel fail to observe safety precautions.

Learn the areas containing high voltage in each piece of equipment.

Be careful not to contact high-voltage connections when installing or operating this equipment.

Before working inside the equipment, turn power off and ground points of high potential before touching them.

Sample B.

Samples of Warning Page Information

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4.2.4 List of effective pages. The front matter of every volume and separately bound part of a volume will have a list of effective pages. In the support volume or part containing the General Information, the list of effective pages will either back up the title page or be a separate page, and will be a complete list, encompassing all volumes and parts thereof. All other volumes and parts will have a list covering only the included contents (see example in MIL-M-24100). An example of an original list of effective pages for an unclassified manual is shown below.

Reproduction for non-military use of the information or illustrations contained in this publication is not permitted. The policy for military use reproduction is established for the Army in AR 380-5, for the Navy and Marine Corps in OPNAVINST 5510.1B, and for the Air Force in Air Force Regulation 205-1.

Insert latest changed pages; dispose of superseded pages in accordance with applicable regulations.

NOTE: On a changed page, the portion of the text affected by the latest change is indicated by a vertical line, or other change symbol, in the outer margin of the page. Changes to illustrations are indicated by miniature pointing hands. Changes to wiring diagrams are indicated by shaded areas.

Total number of pages in this manual is 438, consisting of the following:

Page No.	#Change No.	Page No.	#Change No.	Page No.	#Change No.
VOLUME 1		5-24 Blank	0	5-54 Blank	0
Title	0	5-25	0	5-55	0
A	0	5-26 Blank	0	5-56 Blank	0
i-iv	0	5-27	0	5-57	0
1-0 - 1-6	0	5-28 Blank	0	5-58 Blank	0
2-1 - 2-11	0	5-29	0	5-59	0
3-1 - 3-3	0	5-30 Blank	0	5-60 Blank	0
3-4 Blank	0	5-31	0	5-61	0
3-5 - 3-7	0	5-32 Blank	0	5-62 Blank	0
3-8 Blank	0	5-33	0	5-63	0
3-9 - 3-15	0	5-34 Blank	0	5-64 Blank	0
3-16 Blank	0	5-35	0	5-65	0
3-17 - 3-19	0	5-36 Blank	0	5-66 Blank	0
3-20 Blank	0	5-37	0	5-67	0
3-21 - 3-23	0	5-38 Blank	0	5-68 Blank	0
3-24 Blank	0	5-39	0	5-69	0
3-25 - 3-26	0	5-40 Blank	0	5-70 Blank	0
3-26A	0	5-41	0	6-1 - 6-32	0
3-27	0	5-42 Blank	0		
3-28 Blank	0	5-43	0	VOLUME 2	
3-29	0	5-44 Blank	0	Title	0
3-30 Blank	0	5-45	0	2-1 - 2-6	0
3-31 Blank	0	5-46 Blank	0	2-7	0
3-32 - 3-53	0	5-47	0	2-8	0
4-1 - 4-2	0	5-48 Blank	0	2-9 - 2-15	0
5-1 - 5-19	0	5-49	0	2-16 Blank	0
5-20 Blank	0	5-50 Blank	0	2-17 - 2-18	0
5-21	0	5-51	0	2-19	0
5-22 Blank	0	5-52 Blank	0		
5-23	0	5-53	0		

*Zero in this column indicates an original page.

Volume 1 List of Effective Pages

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4.2.5 Promulgation page. An agency of the procuring activity will supply this page, when required. This should be requested just prior to the printing of the manual.

4.2.6 Change record. The format for the change record page is shown below.

CLASSIFICATION

PUBLICATION NUMBER

RECORD OF CHANGES

CHANGE NO.	DATE	TITLE OR BRIEF DESCRIPTION	ENTERED BY

CLASSIFICATION

Change Record Format

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4.2.7 Table of contents and list of illustrations. The format and content rules for these two items follow the requirements specified in MIL-M-38784. However, the tabulations only cover the contents of the support volume. Format examples of the table of contents and list of illustrations are shown below.

TABLE OF CONTENTS

Section		Page
	VOLUME 1	
	LIST OF ILLUSTRATIONS	ii
	FUNCTIONAL INDEX	iii
	FOREWORD	v
	SAFETY SUMMARY	vii
1	GENERAL INFORMATION	1-1
	1-1. Introduction	1-1
	1-5. Equipment Description	1-3
	1-7. Unit 1, Indicator Transmitter	1-5
	1-8. Unit 2, Remote Control	1-7
	1-9. Unit 3, Rodmeter	1-9
	1-10. Unit 4, Sea Valve	1-11
	1-11. Relationship of Units	1-13
	1-13. Reference Data	1-13
	1-15. Equipment, Accessories, and Documents Supplied	1-15
	1-17. Equipment and Publications Required but not Supplied	1-16
2	OPERATION	1-19
	2-1. Introduction	1-19

Single Column Table of Contents

LIST OF ILLUSTRATIONS

Number	Title	Page
1	Electromagnetic Underwater Log Equipment	1-2
2	Unit 1, Indicator Transmitter	1-4
3	Unit 2, Remote Control	1-6
4	Unit 3, Rodmeter	1-8
5	Unit 4, Sea Valve	1-10
6	Relationship of Units	1-12
7	Controls and Indicators	1-22
8	Induced Voltage in Plane of Water	1-36
9	Performance Test Set-up	1-45

Single Column List of Illustrations

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4.2.8 Functional index. The functional index is a tabulation of the information contained in the T/R volume. The text and example in MIL-M-24100 are self-explanatory. Other matrix-type formats such as the ones shown below, are acceptable.

FUNCTIONAL DATA

	PAGE NO.		
	DIAGRAM	TEXT	MDC
FAMILY TREE	7		
INTERCONNECTING CABLING/PIPING DIAGRAM	8		
ALIGNMENT PROCEDURES		9	
TURN-ON/CHECK-OUT CHART	11		
OVERALL INFORMATION	15	14	17
INTERMEDIATE INFORMATION	19	18	18
AC POWER DISTRIBUTION	21	20	20
DC POWER DISTRIBUTION	23	22	22
TRANSMIT FUNCTION	25	24	26
RECEIVE-RECORD FUNCTION	29	28	27
DIGITAL DISPLAY FUNCTION	31	30	32, 33
REMOTE INDICATOR FUNCTION	35	34	34

HARDWARE DATA

		PAGE NO.				
REF DESIG	NAME	PARTS DATA	TEXT	SCHEMATIC	MAINTENANCE DATA	WIRING DIAGRAM
1						
1A1						
1A2						
1A3						
1A3A1						
1A3A2						
1A4						
2						

NOTE: OTHER INDEXING SYSTEMS WHICH BETTER SUIT THE PARTICULAR EQUIPMENT SITUATION ARE ACCEPTABLE

Functional Index

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4.2.9 Foreword. This information is intended to answer the what, why, when, where, and how questions concerning the manual and its use. It should describe the data contained in the manual, and the order and relationship of its presentation. Detailed instructions concerning use of the data should also be included. Data use instructions will also be included in the front matter for volume 2.

4.2.10 Safety summary. The safety summary should be a compilation of all general precautions, warnings, and cautions extracted from a particular volume or part. It is included in the front matter of the volume or part to which it pertains. Cautions and warnings should be rephrased if they are hard to understand when read out of context. A typical example is shown below.

SAFETY SUMMARY

The following are general safety precautions that are not related to any specific procedures and therefore do not appear elsewhere in this publication. These are recommended precautions that personnel must understand and apply during many phases of operation and maintenance.

KEEP AWAY FROM LIVE CIRCUITS

Operating personnel must at all times observe all safety regulations. Do not replace components or make adjustments inside the equipment with the high voltage supply turned on. Under certain conditions, dangerous potentials may exist when the power control is in the off position, due to charges retained by capacitors. To avoid casualties, always remove power and discharge and ground a circuit before touching it.

DO NOT SERVICE OR ADJUST ALONE

Under no circumstances should any person reach into or enter the enclosure for the purpose of servicing or adjusting the equipment except in the presence of someone who is capable of rendering aid.

RESUSCITATION

Personnel working with or near high voltages should be familiar with modern methods of resuscitation. Such information may be obtained from the Bureau of Medicine and Surgery.

The following warning appears in the text in this volume, and is repeated here for emphasis.

WARNING

High voltages capable of causing death are used in this equipment. Use extreme caution when servicing either the power supplies or their load components.

Example of Safety Summary

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4.3 Section 1. General information. The general information provides a summary of the equipment (its functions, characteristics, and operation) for command level, supervisory personnel, and other users. The information required in this section is specified in detail in MIL-M-24100.

4.4 Section 2. Operation. The information in this section covers operation of the equipment in all modes and under all conditions. It is important that the information convey to the operator sufficient knowledge about equipment performance so he can recognize whether there are:

(1) Hardware problems
(2) Improper manipulations or settings of controls
(3) Environmental considerations affecting equipment operation.

A well-developed Operation Section will greatly increase equipment "up-time" and reduce apparent need for "service" and "down-time". Such data will reduce two types of operator errors: the belief that the equipment is not working when it really is, and the belief that it is working when it really is not. The formats to be used are detailed in MIL-M-24100.

4.4.1 Operator data table. The operator data table must provide a concise picture of controls and indicators, and describe their position or indications during normal operation. The information should be sufficient to enable a manual user, untrained or inexperienced at that particular station, to operate the equipment in an emergency.

4.4.2 Operating procedures. These procedures should be complete, step-by-step instructions necessary for operation of the equipment in all modes and under all conditions. They should be designed for use in classroom and on-the-job training, and also serve as a checklist or supplemental data for experienced personnel. When operating procedures have a higher security classification than the equipment itself, they will normally be contained in a separate publication or volume. Only abbreviated procedures, compatible with the manual classification, will then be included in this section.

4.4.3 Turn-on/check-out procedures. These instructions direct the operator in determining the operational readiness of the equipment. They can be presented in paragraph format, or as a chart similar or identical to the one in volume 2. With either format, the steps in the support volume procedures must be compatible with the steps in the T/R volume chart.

4.4.4 Operator's maintenance actions and schedules. These instructions should direct the operator in completing certain preventive maintenance procedures and performing minor repairs or adjustments. The level of operator action is specified by the adopted operating philosophy of the procuring activity or other cognizant agency. Presentation of the information can be in either tabular or paragraph format.

4.5 Section 3. Theory of operation. This section is intended to explain the basic physical laws involved in equipment operation, and the theory of their application to the designated task. Equipment capabilities, and their impact on overall system or activity mission effectiveness, should also be discussed.

Before preparing this section, it would be advantageous to consider the difference between equipment operation and equipment function as it applies to FOMM. In essence, operation is "what", and function is "how". Text in volume 2 deals predominantly with "how": how entities work together to perform functions; how functions interrelate to perform the overall function. The theory of operation discusses what is accomplished by the equipment, and the "concepts" of how, rather than the mechanics. The preceding statement is not contradictory. To say that a piece of equipment locates metal by detecting variations in the earth's magnetic field describes an application of theory. The circuitry required for its implementation is quite another matter. This does not preclude description of innovative circuits or unusual circuit applications when an understanding of their operation facilitates overall comprehension. However, these descriptions and illustrations should cover only conceptual applications.

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4.6 Section 4. Scheduled maintenance. The purpose of this information, as with any scheduled maintenance plan, is to prevent equipment failure before it occurs. The requirements are outlined in MIL-M-24100. Planned maintenance for the same generic family of equipment, and the design criteria for the equipment being documented, will establish the guidelines to be considered.

4.6.1 Introduction. This should be a brief paragraph describing the scheduled maintenance information, its purpose and arrangement. For Navy manuals, the following statement shall be included: "Where conflict arises between the procedures contained on the Planned Maintenance System 'Maintenance Requirement Cards (MRC)' for this equipment and those in this manual, the procedures on the MRC's shall take precedence".

4.6.2 Scheduled maintenance action index. This is a quick reference for maintenance actions, to be performed at designated intervals. Periodicity will have to be determined from generic information and design criteria.

4.6.3 Preventive maintenance procedures. The intent of these instructions is to prevent equipment failure by regular inspection and care of known or suspected problem areas. Prescribed actions should be the result of past experiences with similar equipment, and design criteria for the equipment being documented. They include methods to retard part degradation, and perform replacement procedures for those items where deterioration or depletion is unavoidable.

4.6.4 Scheduled performance tests. The object of these tests is to detect deficiencies in equipment operation before they become major problems. This permits corrective actions to be taken during periods of planned equipment inactivity, and thus reduces unscheduled down-time. As with other scheduled procedures, design criteria and records concerning similar equipment will play a major role as a source of information.

4.6.5 Plan for maintenance of mechanical components. These instructions cover mechanical parts which, regardless of normal preventive maintenance procedures (cleaning, lubrication, etc.), will eventually require overhaul or repair. The representative considerations covered in MIL-M-24100 will be used to determine the actions required and the frequency of these actions.

4.7 Section 5, Installation. The source of the installation data will be engineering documents, but it must be arranged to comply with MIL-M-24100 formatting standards. The content of this section should be sufficiently detailed so as to allow an experienced installer to accomplish the task. Information requirements, from unpacking to final installation verification check are covered in detail in MIL-M-24100.

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4.8 Section 6, Parts list. Parts data, when required in the support volume, is intended to be a separate or separable document for logistics and supply personnel. If the manual is prepared for the Navy, Marines, or Coast Guard, the parts information covered is in MIL-M-24100. Separate parts lists formats for other agencies are covered by documents specified in the contract.

Regardless of the volume 1 requirements, there will always be parts lists included with the hardware data packages in volume 2. In addition to contractual requirements, basic format differences exist between electronic and electrical, mechanical, hydraulic, and pneumatic parts lists.

4.8.1 Introduction. The introduction to a separate parts list names the equipment covered, explains the arrangement of the data provided, its content, and its use.

4.8.2 List of major components or major units. This is a tabular listing of the major replaceable items comprising the documented equipment. Electrical, mechanical, hydraulic or pneumatic equipment requires a major components list, while electronic equipment is covered by a list of major units and assemblies. It is also quite possible that the subject equipment be a combination of electronic and one or more of the other types. When this occurs, the electronic format can be used listing items without reference designators first. Items without designators are listed in alphabetical order, as shown below.

TABLE 7-1. LIST OF MAJOR UNITS

UNIT NUMBER	NOMENCLATURE		PAGE NUMBER
	NAME	DESIGNATION	
	Cabinet Assembly	14D1853	7-5
	Heat Transfer and Upper Assembly	30E3749	7-7
	Top Assembly	46E915	7-4
A101	Module Assembly, Single Phase	46E906-1	7-13
A101A1	Half Bridge Assembly	30E3738-1	7-15
A101A2	Half Bridge Assembly	30E3738-2	7-15
A102	Module Assembly, Single Phase	46E906-2	7-13
A102A1	Half Bridge Assembly	30E3738-1	7-15
A102A2	Half Bridge Assembly	30E3738-2	7-15
A103	Module Assembly, Single Phase	46E906-3	7-13
A103A1	Half Bridge Assembly	30E3738-1	7-15
A103A2	Half Bridge Assembly	30E3738-2	7-15
A104	Right Hand Extension Assembly	30E3783	7-31
A105	Left Hand Extension Assembly	30E3782	7-35
A106	High Voltage Assembly	30E3865	7-39
A107	Control Console Assembly	30E3753	7-43
A108	Voltage Sensing Assembly	30E3866	7-90

List of Major Units

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4.8.3 Parts list requirements. Each parts list should contain complete ordering data for every replaceable item on the documented assembly or subassembly. Format differences exist between electronic and other types of equipment, but the information and depth of coverage required is basically the same.

In column 1 of an electronic equipment parts list, the entire reference designator for each part must be shown. For example, 1A2A1AR15 would be the listed reference designator for component AR15 on subassembly A1 on assembly A2 of Unit 1. When a single parts list is used for identical assemblies (e.g., 1A2A1/1A2A2/1A2A3), reference designators become quite lengthy (e.g., 1A2A1AR1, 1A2A2AR1/1A2A3AR1) but still must be shown in their entirety. Items without reference designators are listed first, in alphabetical order, as shown below.

TABLE 7-6. HALF BRIDGE ASSEMBLY A101A1/A101A2/A102A1/A102A2/A103A1/A103A2 (Continued)

REFERENCE DESIGNATION	NOTES	NAME AND DESCRIPTION	FIGURE NUMBER (ITEM)
		SEPARATOR: 18 5/8" x 2 7/8" x 0.06 THICK ALUMINUM ALLOY 6061-T6, mfr 90911 part no. 30E3738-94 (common item No. 54)	7-4 (32)
		SPACER: mfr 06540 part no. 8872-A-2520 (common item No. 58) (connecting parts) EB(1)	7-4 (33)
		SPACER, FRONT PANEL ASSEMBLY: 1/4" BAR ALUMINUM ALLOY 6061-T6, mfr 90911 part no. 25B4849 (common item No. 59)	7-4 (34)
		STANDOFF: 5/8 RND x 1 1/4 LONG (5/16-18) mfr 06540 part no. 8878-A-3118 (common item No. 60) (connecting parts) EJ(1), DX(1)	7-4 (35)
		STUD: 1/4-20 x 1 CADMIUM PLATED STEEL, mfr 90911 part no. 30E3738-98 (common item No. 62)	7-4 (36)
		STUD: 5/16-18 UNC x 1 1/4 CADMIUM PLATED STEEL per QQ-P-416, mfr 90911 part no. 30E3757-98 (common item no. 61) (connecting parts) EF(1), R(1), S(1)	7-4 (37)
		TERMINAL, INSULATED: mfr 88245 part no. 1423 (common item No. 64) (connecting parts) O(1)	7-4 (38)
		TERMINAL, FLAG: 1/4" HOLE, NO. 4 WIRE, mfr 59730 part no. F871 (common item No. 63) (connecting parts) AL(1), AQ(1), BY(2), AN(1)	7-4 (39)
		WATER MANIFOLD, HALF BRIDGE: mfr 90911 part no. 25D5463 (common item No. 71) (connecting parts) L(2), DI(2), M(2)	7-4 (40)
A101A1C1A/ A101A2C1A/ A102A1C1A/ A102A2C1A/ A103A1C1A/ A103A2C1A		CAPACITOR: 1200-1250UF, 450WVDC, mfr 90911 part no. 22C1047 (common item No. 17) (connecting parts) V(1)	7-4 (41)

Parts List

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4.8.4 Additional listings. Three additional items should be generated in the process of preparing the parts list. They are the lists of common items, the list of attaching hardware, and the list of manufacturers.

The list of common items provides a separate listing of multiple-use items for use by supply personnel.

The list of attaching hardware makes unnecessary the repetition of parts data for small mechanical parts (screws, washers, nuts, etc.). Identification letter codes enable their inclusion with the items they attach without greatly increasing the bulk of the parts list.

The list of manufacturers should contain the names and addresses for all the manufacturers whose code symbols are used in the parts list. This eliminates the need for referencing code number handbooks when ordering parts.

4.8.5 Parts location illustrations. Requirements for parts location illustrations are explained in detail in MIL-M-24100.

5. PRODUCTION OF DELIVERABLE DATA

5.1 Book plan (dummy manual). The dummy manual illustrates the contractor's intended implementation of FOMM formatting techniques to provide equipment documentation in accordance with MIL-M-24100. It provides a view of the scope and level of coverage of the proposed manual. The dummy manual will be used by the procuring activity to ensure that the coverage proposed by the contractor will be adequate. It also provides the contractor with a useful bidding and scheduling tool. In multi-volume manuals, the support volume book plan will indicate the arrangement of data to the paragraph level, in accordance with MIL-M-38784. The T/R volume dummy manual should be a page-by-page plan for the entire final manual. A rough sketch of each page of the final manual should be prepared, indicating approximate space allocations and arrangement of the required FOMM data. Generalized statements, such as "Seventeen similar pages to follow" or, "This is a sample of a hardware data package, there will be twenty such data packages", are not sufficient. The dummy manual may be prepared on 8½ x 11 inch paper, regardless of final page size. Figure 26 is an example from an acceptable dummy manual prepared for an 11 x 27 inch T/R volume. There is obviously some speculation involved with space allocation. To do this with any accuracy, the writer must become familiar with the engineering source data. Once actual preparation of the manual has begun, the dummy manual must be updated to reflect the real manual.

5.2 QA plan. A quality assurance (QA) plan is necessary to assure that the manual accurately reflects the equipment and contains all required data items. The contractor must provide the procuring activity with proof of an adequate quality program before any work can be done on the manual. Documentation of the items noted below is required. They are covered in detail in MIL-M-24100.

(a) Organization: designation of qualified personnel responsible for quality control.
(b) Source data control: assurance that the latest engineering source data is used in the generation of the manual and that authorized changes are included.
(c) Records and checklists: plan for documentation of the required quality checks.
(d) In-process reviews: scheduling of in-process inspections during the preparation of the manual.
(e) Validation plan: a record of the check of every data item against the source data or equipment.
(f) Validation: demonstration of the accuracy of all data items by actual performance on the equipment.

5.3 Original artwork. Original artwork for volume 1 is prepared in accordance with MIL-M-38784. However, all illustrations must be reproducible, either by photocopy or ozalid process. Artwork for volume 2 required special techniques for preparation and control.

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5.3.1 Drawing control. Due to the fact that drawings must remain accessible for reference, and for modification (up until the freeze date of the manual), a comprehensive drawing control system is an absolute necessity. Information concerning the immediate location and stage of completion for each drawing must be readily available. Two items suggested for this purpose are a drawing status report and a drawing control sheet.

5.3.1.1 The status report enables the writer and the writer's management to keep track of the progress of individual drawings. Photocopies of the report can also be used to indicate program status in the periodic reports required by the purchasing activity.

5.3.1.2 The drawing control sheet is initiated by the engineer/writer who generates the first rough layout of the drawing. As it proceeds through stages of development, those working on the drawing make note of the task performed and the time expended for that task. The drawing control sheet is used to update the status report, and to keep track of total expended hours.

5.3.2 Writing hints. The functional layout originated by the writer should be done on the same size paper as that used by the illustrator. This will force him to keep in mind the page size limitations with which the illustrator must contend. Upsize art image area can vary within certain bounds. This is limited by the amount the drawing must be reduced to fit on a manual page, and still meet the minimum type size requirements specified in MIL-M-24100. The writer can also aid the illustrator by keeping in mind the microfilm requirements, and arranging his rough layout so that it breaks cleanly into three panels. Functional blocking and hardware boundaries must be clearly defined. A technique found successful for this purpose is the use of tic marks and color coding. Tic marks are provided at each 90 degree angle of the boundaries, and are linked by colored lines indicating the functional or hardware level they contain. The color code below is suggested.

Functional	Hardware
1st level - light blue	1st level - yellow
2nd level - dark blue	2nd level - orange
3rd level - light green	3rd level - red
4th level - dark green	4th level - brown
	5th level - terra cotta

An in-house checklist should be developed for use by the writer to ensure that all required data is included on the rough drawing before its submission to the art department.

5.3.3 Drafting hints. The illustrator, on the other hand, should be made aware that the submitted layout must be adhered to as strictly as possible. Any changes necessitated by space limitations, or to meet MIL-M-24100 microfilm panel requirements, must be discussed with the writer so as not to disrupt the functionality of the drawing. Of course, as the illustrator becomes more knowledgeable in FOMM techniques and requirements, he should be able to make minor layout adjustments without going to the writer each time a minor problem develops. A copy of the illustrator's (usually pencil) layout should be returned to the writer for comparison against his input. The annotated copy is then returned to the illustrator, and the pencil layout corrected. If of sufficient quality, a copy of the pencil layout may be used in the review manuscript (see MIL-M-24100). The final drawing should be prepared on a stable-based, translucent material. The usual method of preparation is to lay the sheet of translucent material over the pencil drawing, allowing the person performing the inking to trace over the pencil lines. In this way, a less experienced illustrator can do the inking, freeing the more experienced for the more difficult original layouts. Acceptable transfer and rub-down symbols should be used to save time and ensure consistency. Hand lettering on final drawings is not permitted. One technique is to type the required information on a clear, sticky-back material, which is cut and appropriately positioned on the drawing by the ink and paste-up illustrator. The drawing is then validated by a line for line check against the engineering source data.

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5.4 Review manuscript. The review manuscript is an approximately final-sized copy of validated manual pages, submitted to the procuring activity before the final manual is produced. Volume 2 copies should be reproduced by the ozalid or equivalent process. Volume 1 copies should be reproduced by the most economical means. The review manuscript contains all the information that will appear in the final manual except for color shading. Hardware boundaries are indicated using the dot-dash method as illustrated by the Overall Function Diagram (Review Manuscript) example in MIL-M-24100. Dotted lines are used to outline functional boundaries. These dot-dash and dotted line boundaries can be added to the reduced size originals before copies are made so they won't appear on the original artwork.

5.4.1 Validation. Validation is that action performed by the contractor to ascertain that the diagrams, procedures, text, operation and all other data of the review manuscript agreeably conform to the dictates of the specification and design of the equipment. Such validation shall be attested to in writing by an officer of the contractor. It should be noted that it would be prudent to invite the reviewing agent (along with the prime contractor, if any) to observe the validation.

5.5 Negatives, masks, and screens. Tint screens are used by the printer to produce the shading that delineates functional and hardware boundaries in the final FOMM manual. After decisions concerning the level of shading and the placement of the various shades were made by the writer as described in paragraph 5.3.2;and the artwork is prepared as described in paragraph 5.3.4, the remaining steps required for the production of the tint screens are:

(a) Make final-size negative.
(b) Cut amberlith masks.
(c) Use masks to make composite tint screens.

5.5.1 Negatives. Final-size negatives for all pages requiring shading are necessary for the generation of the tint screens. It is important that the same negative (or contact reversal) that is used make the masks is used to make the printing plate. If different negatives are used for each task, extremely small variations in reduction can cause registration problems between the line work and the shading.

5.5.2 Masks. The masks are used to restrict light to various areas of the raw film that will eventually become the composite tint screen for a specific page. One mask must be prepared for each shade of blue and each shade of grey appearing on any manual page. The materials that can be used for masking are described in MIL-M-24100. The masking material is laid over the final size negative (or contact positive) of the page on a light table. Using the tic marks, the areas to be indicated by one particular level of blue (or grey) are outlined with a knife or scribe and peeled or "stripped" away, leaving the rest of the mask intact. If a mistake is made, a portion of previously peeled-away masking material can be repositioned over the uncovered area. This process is usually started at the lowest level (lightest) shade, that mask removed from the drawing on the light table, and the next level cut. Each mask must be identified as to the page, color, and shade for which it is to be used. The method for re-registering the masks over the drawing for exposure must be precise, because tic marks are opaqued before printing. One method of registering masks is to hole-punch the negative and the appropriate number of sheets of masking material as a package, before the scribing and stripping process. The holes, and the registration key pins over which the sheets are placed for stripping and screening must be in very close tolerance to preclude slippage. Remember that the grey shades cannot overprint the blue shades. In order to meet this requirement, areas to be printed blue must remain opaque on the grey masks.

5.5.3 Screens. Many techniques are available for the preparation of tint screens. These can be divided into two general groups: through the lens, and contact exposure. Through the lens techniques are common to the industry and, as such, will not be discussed here. They will produce adequate results, but are not cost-competitive with contact screening. The following suggested contact screening technique will accomplish the task with a minimum of expense. The method involves the use of lithographic film, a point light source, a 133 line-per-inch clear dot tone screen, the masks described in the previous paragraph, and a vacuum frame. Multiple lamps are strategically placed over a vacuum frame. The master clear dot screen is placed on the raw film, emulsion side up, not in contact with the film. By selecting the light or cluster of lights and an appropriate mask, the clear dot will be undercut. Since light rays can

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penetrate only the clear dot, the only silver salts on the film that can be affected are those hit by the light rays that go through the dots (undercutting or through the clear area of the screen). Overexposure will not change the resulting dot size. A non-infectious type developer is used to leave the unexposed silver salts unaffected by the development process. Accordingly, time, temperature, and concentration aspects are no longer items of significance. A separate exposure through a different mask is required for each level of shading. Some art groups feel they can produce the shaded areas by use of pre-screened sticky-back material, or by splicing together pieces of tone screens of varying percentages. It is possible to use this method. However, during the printing process, false lines may be created at the junction of the shaded areas due to dots touching one another. Such lines will be in competition with actual data lines, and would be both undesirable and unacceptable.

5.6 Preliminary manual. A preliminary manual may be required contractually, but usually they are produced only when there is insufficient time to meet the requirements for a final manual before the delivery of the equipment being documented. A preliminary manual contains the same information and grey shading as the final manual, but the requirement for blue shading is waived. Functional entities are contained in white areas, within the grey hardware shades. Printing of a preliminary manual is usually a Government Printing Office (GPO) task. The contractor is required to supply GPO with the line negatives, masks, and grey screens for the manual. The contractor is still required to supply a complete, final manual. This supersedes the preliminary manual, which must be updated to accurately represent production equipment.

5.7 Final manual. Printing and binding of a final manual is primarily a GPO function. This normally includes photography of camera-ready copy, supplied by the manual contractor, to produce photolithographic negatives. However, negatives must be made to produce masks and tint screens which are a required deliverable item for FOMM when specified in the contract. The printer is responsible for holding the dot sizes in the final manual to those required by specification. Collation sheets, in accordance with MIL-P-38790 or other instructions, are also required.

5.8 Technicians pocket manual. The technicians pocket manual is a photo reduction of the T/R volume. A new publication number is assigned for the cover. Because of its smaller size, new screen composite negatives must be made using a 200 line per inch master screen. Semigloss paper must be used for printing the pocket manual to ensure accurate reproduction of the smaller dots and characters. The Major Function Diagram (Digital) example in MIL-M-24100 is printed on a sample of this paper. If a Technicians Pocket Manual is required by contract, page reductions used for producing the original T/R volume cannot be as great,due to the smaller manual typography specifications.

6. ESTIMATING AND BIDDING

6.1 Introduction. Scheduling the production of a FOMM publication - and ultimately the cost of the program - is greatly influenced by the time allowed for the program,and the organizational structure of the company performing the work. For an estimate to be valid, the estimator must decide on the number of writers to be used and their qualifications with respect to the assignment of tasks. It must also be decided how much of the graphic layout responsibility is to be in the hands of the writer, and how effective communications between artist and writer is to be accomplished. Finally, decisions must be made on the scheduling of editing and quality control actions,and whether these tasks will be performed using copies of original artwork or reductions to final manual size. It is expected that decisions on the above will be made to most efficiently use the resources available. Therefore, there will be differences in the method of operation from one organization to another, and so the information given here is general in nature. By the time an organization completes its first FOMM job, appropriate estimating and bidding techniques for the organization should have been developed.

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6.2 Contract requirements. The specification lists some 16 items which can be specified as deliverables by the contracting agency. Each of these items, if required, will be found on the appropriate contract data requirements list. Also included in the list will be statements on quantities, delivery dates, applicable specifications, and modifying/clarifying data item description references. All of this information must be understood in order to prepare a valid bid. In some cases, it may be advisable to ask for changes in the requirements. As an example, a procuring activity more familiar with a standard technical manual (as defined in MIL-M-38784) may establish an unrealistic delivery date for the review manuscript. In this case, it is suggested that the estimator request the scheduling of in-process reviews in lieu of an early review manuscript. The reason for this is that changes after submission of the review manuscript can be very costly.

6.3 Preparing a book plan. A dummy manual (book plan) should always be a deliverable data item. Additionally, a dummy manual is the best source for estimating information; thus for the T/R volume, a dummy manual should be prepared as part of the bidding process. For the support volume, a dummy manual is not as important, and the standard in-house bidding process may be followed, keeping in mind that the scope of the volume (compared to a conventional manual) is very limited. When outlining a dummy manual, plan in terms of the basic packages of data. The planning process can begin with an analysis of the number of distinct hardware divisions which require data packages consisting of parts information, text, schematic, MDC, signal specifications, and other maintenance information. After analysis of the hardware data, an estimate of the number of major functions can be made. This estimate is based upon the amount of block diagram information which can be placed on the size of page to be used. If the option is left to the estimator, then it is suggested that the selection of page size be determined by the size needed to produce block diagrams for meaningful functional divisions. It should be kept in mind that, while it is desirable to limit page size to less than the specified maximum, the selection of too small a size will result in more subfunctions and intermediate data packages, with a consequent increase in page units and cost. It is the experience of most estimators that there is a tendency to underestimate the number of major functions which will ultimately be developed. Some of this may be due to the acceptance, for bidding purposes, of the engineer's list of functions; and then the discovery, during manual preparation, that things were not all that simple. One course of action is to consider one major function package for each value of voltage used to power the system; then add a data package for each function defined by engineering; and finally add a few more for items classified as built-in test, failure detection, display, etc. Estimating the pages of overall function information should produce little problem since there is only one overall data package, and its contents are well defined. Estimating the page count for front matter, introduction to the manual, and introduction to the equipment, may be more difficult if one has never seen a FOMM volume. If at all possible, obtain a copy of a published volume. It will be found that since there is some standardization in this material, one volume will serve as a guide for another, and some information can be copied directly or edited.

6.4 Estimating labor. Estimating labor, like preparing the dummy book, should start with consideration of the number of hardware data packages. The number of packages, multiplied by the per hour average for each basic page type, multiplied by the estimated deviation from a standard factor for each of the basic page types, should produce the hours required for that part of the task. These hours, when multiplied by the labor category rate of the personnel to be used, will produce direct cost for that portion of the program. This procedure is then repeated for the upper levels of data, with the estimate for the major function information being subject to the greatest error. The difficulty in estimating costs for major function information is due to the possible error in determining the quantity of major functions, and the fact that the cost of preparing an individual data package tends to increase as a function of the complexity of the block diagram. For the first FOMM bid, it is probably not a good idea to attempt to equate page unit costs for other manuals with that of the proposed FOMM. Normally, the information density of a FOMM is much, much greater than in a conventional manual. It is to be expected that for software, as is true with hardware, a small package costs more per pound than a functionally equivalent large package.

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6.5 Estimating and bidding summary. Material costs for a FOMM program are easier to estimate because of the reduced number of types of data pages. Additionally, the application of the FOMM technique to an equipment with precise data requirements provides both the customer and contractor with a very close approximation of an actual page count. This provides both parties with a realistic view of what is to result as an end product. Also, any minor fluctuations in the end product can be negotiated realistically to the mutual advantage of both customer and contractor. It is unlikely that any book bid to any system will come out exactly as planned. Changes in the engineering concepts of the equipment during the design are unpredictable. Minor changes resulting from handbook decisions to combine pages, or to subfunctionalize pages, are expected and will occur on all but the smallest programs. The advantage of a FOMM manual from a planning or bidding viewpoint is that each data requirement is precise. Thus, the course of decisions on the data packages from the bid to the final product can be reconstructed quite simply to evaluate the changes of requirements that occur. As such, the nebulous nature of both bid and negotiating costs can be eliminated. One problem the bidder may have is estimating the amount of material to be produced for editing, quality control, and review purposes. An effective way of minimizing this cost is to plan the program fully in advance and establish good in-house procedures to ensure quality, and thus minimize rework. Finally, close communications must be maintained with the reviewing activity so as to reach an agreement on the manual contents prior to submission of the review manuscript.

Preparing Activity:
Navy-SH
(Project TMSS-0229)

Custodians:
Army - TM
Navy - SH
Review Activities:
Army - AT, AV, CE, EL, ME, MI, MU, SC, WC
Navy - CG, EC, MC
User Activities:
Navy - OS, YD

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